
Polynomial–Logarithmic Transseries

Arithmetic, Division, Composition, and Series Reversal

A constructive guide centered on the asymptotic inversion
behind Lambert’s W function

From coefficient convolution and reciprocal recurrences to
normal-ordered composition, Bell-polynomial bookkeeping, Newton
reversion, Stirling-number formulas, branch-sensitive Lambert
expansions, and implementable algorithms.

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Abstract

Polynomial–logarithmic transseries are expansions whose coefficients are polynomials in a slow logarithmic variable and whose successive levels are powers of a faster variable. The model example is

$$W(z) = X - \log X + \frac{\log X}{X} + \frac{\log X(\log X - 2)}{2X^2} + \cdots, \quad X = \operatorname{Log} z + 2\pi i k,$$

which inverts $w \mapsto we^w$ on the k -th branch. These expansions sit between ordinary formal power series and the full logarithmic–exponential transseries field. They are elementary enough for explicit coefficient algorithms, yet rich enough to display the essential phenomena of asymptotic scales, logarithmic coefficient growth, failure of naive closure, and compositional inversion.

This article develops a self-contained calculus for such series. It begins with the ring $\mathbb{K}[L]((X^{-1}))$, explains its valuation and asymptotic meaning, and carefully distinguishes it from the larger field $\mathbb{K}(L)((X^{-1}))$. It then treats multiplication, powers, reciprocal and quotient recurrences, differentiation, logarithms and exponentials, regular and near-identity composition, Bell-polynomial coefficient extraction, and formal series reversal. Universal composition and inverse formulas are displayed through fourth order. Newton algorithms are proved to double formal precision. The Lambert W expansion is derived twice—by triangular coefficient matching and by an explicit Stirling-cycle-number formula—and is extended to all complex branches, the real branch W_{-1} , and equations of the form $y + \alpha \log y = X$. A final implementation chapter turns the theory into precise data structures, pseudocode, verification tests, and a compact SymPy prototype.

The emphasis throughout is constructive: every infinite operation is reduced to a finite calculation at a requested truncation order, every closure claim is stated with its hypotheses, and every important formula is accompanied by a residual check.

Reader's guide

The basic prerequisites are calculus, ordinary power series, and comfort with formal algebra. Familiarity with asymptotic notation is helpful but not required. Readers mainly interested in computation may read Chapters 2, 3, 4, 5, 6, and 7 in order, then jump to Chapter 10. Readers interested in the boundary between this elementary calculus and general transseries should also read Chapters 8 and 9.

Two notational cautions are worth stating immediately.

1. The symbol A^{-1} is ambiguous. We write $A^{(-1)} = 1/A$ for a multiplicative inverse and $A^{[-1]}$ for a compositional inverse.
2. The variable L always means $\log X$ unless a chapter explicitly switches to a small variable. It is not an independent variable under composition: changing X changes L as well.

Scope and closure

The phrase *polynomial-logarithmic transseries* is used here for a practical, explicitly computable subclass of power-log transseries. The core class has integer powers of X^{-1} and polynomial coefficients in $L = \log X$. Later chapters allow rational coefficients in L , generalized real exponents, and finitely many iterated logarithms. Full transseries permit much more, including exponentials of large transseries; the present article uses that larger theory as context but does not require it.

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Why polynomial–logarithmic transseries appear

1.1 The first logarithm forced by inversion

Suppose that a large positive quantity Y is related to another large quantity X by

$$Y + \log Y = X. \quad (1.1)$$

If the logarithm were absent, the answer would simply be $Y = X$. Since $\log Y$ is much smaller than Y , a first correction is

$$Y \approx X - \log X.$$

But this is not exact. Substituting it into the left side of Equation (1.1) gives

$$X - \log X + \log(X - \log X) = X + \log\left(1 - \frac{\log X}{X}\right).$$

Expanding the last logarithm produces

$$-\frac{\log X}{X} - \frac{(\log X)^2}{2X^2} - \frac{(\log X)^3}{3X^3} - \dots.$$

The correction to $X - \log X$ therefore involves powers of X^{-1} , and their coefficients involve powers of $\log X$. Continuing the calculation gives

$$\begin{aligned} Y = X - L + \frac{L}{X} + \frac{L(L-2)}{2X^2} + \frac{L(2L^2-9L+6)}{6X^3} \\ + \frac{L(3L^3-22L^2+36L-12)}{12X^4} + \dots, \quad L = \log X. \end{aligned} \quad (1.2)$$

Equation (1.1) is exactly the logarithmic form of the Lambert equation $Ye^Y = z$, with $X = \log z$. Thus Equation (1.2) is the large-argument expansion of Lambert's W function.

Intuition

An inverse expansion remembers every asymptotic scale that was needed to undo the original function. The dominant term Y in $Y + \log Y$ creates the scale X ; the smaller term $\log Y$ creates the scale $\log X$; repeated correction creates powers of X^{-1} carrying polynomial dependence on $\log X$.

1.2 A scale between power series and full transseries

Ordinary power series use one small monomial, say q :

$$a_0 + a_1 q + a_2 q^2 + \cdots.$$

At infinity we may take $q = X^{-1}$. A polynomial–logarithmic series allows the coefficient of each q^n to be a polynomial in the slower variable $L = \log X$:

$$A(X) = \sum_{n \geq n_0} a_n(L) X^{-n}, \quad a_n(L) \in \mathbb{K}[L]. \quad (1.3)$$

Examples include

$$X - L + \frac{L}{X}, \quad 1 + \frac{L^2 - 3L + 1}{X^2}, \quad X^{3/2} \left(1 + \frac{2L - 1}{X} + \cdots \right).$$

This is already a transseries in the modest sense that it combines more than one asymptotic scale. For every fixed integer m ,

$$X^{-1} \prec L^{-m} \prec 1 \prec L^m \prec X \quad (X \rightarrow +\infty),$$

where $f \prec g$ means $f/g \rightarrow 0$. Thus one step in the power of X dominates every finite change in the power of L .

Full logarithmic–exponential transseries allow arbitrary well-based sums of monomials built from powers, exponentials, logarithms, and iteration. They are the right global setting for closure under broad classes of operations [4, 5, 13, 1]. The smaller class (1.3) has a different advantage: each coefficient can be manipulated by finite polynomial algebra, and the truncation order in X^{-1} supplies a simple valuation.

1.3 Four recurring sources

Polynomial–logarithmic expansions arise in several closely related ways.

1.3.1 Asymptotic functional inversion

If

$$f(y) = y + \alpha \log y + \frac{p(\log y)}{y} + \cdots,$$

then its inverse usually has the form

$$f^{[-1]}(x) = x - \alpha \log x + \sum_{n \geq 1} \frac{P_n(\log x)}{x^n}.$$

This includes Lambert W , generalized Lambert equations, asymptotic inverses used in prime-number estimates, and inverses of many exp–log functions. Symbolic algorithms for broader exp–log inverse problems were developed by Salvy, Shackell, Richardson, and van der Hoeven [11, 12, 10].

1.3.2 Implicit equations

Even when no named inverse function is available, an equation such as

$$y + \log y + \frac{\log y}{y} = x$$

can be solved coefficient by coefficient. The formal residual determines one new polynomial at each order. This triangular mechanism is one of the main themes of this article.

1.3.3 Iteration and normal forms

Power–log series also occur in the formal classification of maps near fixed points and in Dulac expansions associated with planar dynamical systems. Composition and compositional inversion are then intrinsic rather than auxiliary operations. In this literature one often works near $x = 0^+$, with monomials $x^\alpha (\log(1/x))^m$, and sometimes must admit iterated logarithms [8, 7, 9].

1.3.4 Asymptotic solutions of differential equations

Differential equations whose dominant balance is algebraic or logarithmic can produce coefficients that are polynomial in $\log X$. Resonance is a common mechanism: when the linear operator that normally determines a coefficient loses invertibility on a particular exponent, a logarithm enters. Although our main focus is algebra and inversion, the differentiation rules developed below are exactly those used in such calculations.

1.4 Formal equality and analytic meaning

A formal series is an algebraic object. The statement

$$A \hat{=} B$$

will mean that the two formal coefficient arrays agree, not that two infinite numerical sums converge. We write

$$A = B + O_q(q^N), \quad q = X^{-1},$$

when the coefficients of q^n agree for every $n < N$. This is a *formal* big-O notation.

An actual function $f(X)$ has a polynomial–logarithmic asymptotic expansion if its successive remainders are controlled in the corresponding ordered scale. One convenient formulation is the following. Suppose $a_n \in \mathbb{K}[L]$ and let $d_N = \max_{n \leq N} \deg a_n$. We say that

$$f(X) \sim \sum_{n \geq n_0} a_n (\log X) X^{-n}$$

if for every $N \geq n_0$ there exists an integer M_N such that

$$f(X) - \sum_{n=n_0}^N a_n (\log X) X^{-n} = \mathcal{O}\left(X^{-N-1} (\log X)^{M_N}\right) \quad (1.4)$$

through the sector or ray under consideration. Stronger definitions enumerate all individual monomials $X^{-n} L^m$, but Equation (1.4) is well suited to blockwise computation.

A common trap

Formal correctness and analytic validity are separate questions. A formal inverse can exist even when the corresponding analytic branch is absent on a given domain. Conversely, an analytic inverse may have different formal expansions in different sectors because logarithm branches change. Our algebraic operations are branch-neutral until a specific analytic interpretation is imposed.

1.5 The two-variable illusion

It is tempting to regard $q = X^{-1}$ and $L = \log X$ as independent formal variables. For addition and multiplication this causes no harm. Under differentiation or composition it is wrong:

$$\frac{dL}{dX} = \frac{1}{X} = q, \quad \log G(X) \neq L \quad \text{unless } G(X) = X.$$

The relation between the fast and slow variables is the source of the shifted differential operators that will appear later. In particular,

$$\frac{d}{dX} (a(L) X^{-n}) = (a'(L) - na(L)) X^{-n-1}. \quad (1.5)$$

The same coupling is responsible for the derivative terms in all universal composition and reversal formulas.

1.6 A roadmap of the algebra

The main operations have different closure behavior.

Operation	Typical input hypothesis	Natural output class
Addition, multiplication	$\mathbb{K}[L]((q))$	$\mathbb{K}[L]((q))$
Differentiation	$\mathbb{K}[L]((q))$	$q\mathbb{K}[L]((q))$ up to the leading power
Reciprocal	leading coefficient a nonzero constant	$\mathbb{K}[L]((q))$
General reciprocal	leading coefficient a nonzero polynomial	$\mathbb{K}(L)((q))$
Near-identity composition	inner series $X + o(X)$ of controlled form	usually $\mathbb{K}[L]((q))$
General composition	inner leading term contains L^β	may require $\log L$
Near-identity reversal	$F(X) = X + o(X)$ with polynomial–log blocks	$\mathbb{K}[L]((q))$

The word “usually” in the composition row matters. Closure is proved for the specific classes used in this article; unrestricted formal substitution can fail even in much larger transseries settings unless support conditions are imposed.

Chapter summary

Polynomial–logarithmic transseries arise naturally when a leading algebraic scale is corrected by a logarithm and then inverted or composed. The basic normal form is a Laurent series in $q = X^{-1}$ with coefficients in $\mathbb{K}[L]$, where $L = \log X$. Formal truncation is q -adic, while analytic meaning requires a separate remainder statement. The variables q and L are coupled under differentiation and composition.

Coordinates, rings, valuations, and normal forms

2.1 The large-variable coordinate system

Throughout the core chapters, X tends to infinity in a region where a branch of $\log X$ has been fixed. Set

$$q = X^{-1}, \quad L = \log X. \quad (2.1)$$

The coefficient field $\mathbb{K}$ may be $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$. For symbolic work it can also be a rational function field in external parameters.

Definition 2.1 (Polynomial–logarithmic Laurent ring). The polynomial–logarithmic Laurent ring at infinity is

$$\mathcal{P}_{\mathbb{K}} = \mathbb{K}[L]((q)) = \left\{ \sum_{n \geq n_0} a_n(L) q^n : n_0 \in \mathbb{Z}, a_n \in \mathbb{K}[L] \right\}. \quad (2.2)$$

Only finitely many negative powers of q are allowed, but infinitely many nonnegative powers may occur.

Thus $X - L + L/X$ is represented as $q^{-1} - L + Lq$. The notation $\mathbb{K}[L]((q))$ emphasizes that the outer formal-series variable is q , while L belongs to the coefficient ring.

2.2 Valuation and truncation

For nonzero

$$A = \sum_{n \geq n_0} a_n(L) q^n, \quad a_{n_0} \neq 0,$$

define

$$\text{val}_q(A) = n_0, \quad \text{val}_q(0) = +\infty. \quad (2.3)$$

The q -adic valuation satisfies

$$\text{val}_q(AB) = \text{val}_q(A) + \text{val}_q(B), \quad \text{val}_q(A + B) \geq \min\{\text{val}_q(A), \text{val}_q(B)\}.$$

The congruence

$$A \equiv B \pmod{q^N}$$

means $\text{val}_q(A - B) \geq N$. The truncation operator is

$$[A]_{<N} = \sum_{n_0 \leq n < N} a_n(L)q^n. \quad (2.4)$$

Every algorithm below takes N as input and performs only finitely many coefficient operations to compute $[A]_{<N}$.

Proposition 2.1 (q-adic completeness). *Let (A_j) be a sequence in $\mathcal{P}_{\mathbb{K}}$ such that for every N , the truncations $[A_j]_{<N}$ eventually stabilize. Then there is a unique $A \in \mathcal{P}_{\mathbb{K}}$ whose truncation at every order is that stable value.*

Proof. For each integer n , choose $N > n$. Eventual stabilization modulo q^N fixes a unique coefficient $a_n(L)$. These coefficients are compatible as N varies. The resulting formal Laurent series has a lower bound on its support because the sequence is already Laurent-bounded at some initial precision. Uniqueness follows coefficientwise. $\square$

This elementary completeness is what makes Newton iteration a formal algorithm: one does not sum numerical values; one stabilizes more and more coefficients.

2.3 The internal logarithmic order

Within a fixed q-block, powers of L are ordered by degree. For monomials $q^n L^m = X^{-n}(\log X)^m$, the asymptotic order at $+\infty$ is lexicographic:

$$q^{n_1} L^{m_1} \succ q^{n_2} L^{m_2} \iff \begin{cases} n_1 < n_2, & \text{or} \\ n_1 = n_2 \text{ and } m_1 > m_2. \end{cases} \quad (2.5)$$

Indeed, every power of L is smaller than every positive power of X . The block notation $a_n(L)q^n$ deliberately groups finitely many adjacent monomials.

Cancellation can lower the degree inside a block or remove the block entirely. For example,

$$(L^2 + L)q^3 + (-L^2 + 2)q^3 = (L + 2)q^3.$$

An implementation must normalize polynomial coefficients before deciding whether a q-block vanishes.

2.4 Why the polynomial ring is not a field

The coefficient ring $\mathbb{K}[L]$ has only the nonzero constants as units. Hence $\mathcal{P}_{\mathbb{K}}$ is a ring but not a field.

Proposition 2.2 (Units in $\mathbb{K}[L]((q))$). *Let*

$$A = q^r (a_0(L) + a_1(L)q + a_2(L)q^2 + \cdots), \quad a_0 \neq 0.$$

Then A has a reciprocal in $\mathcal{P}_{\mathbb{K}}$ if and only if $a_0(L)$ is a nonzero constant.

Proof. The monomial q^r is invertible, so it suffices to consider $r = 0$. If $AB = 1$, the constant q -coefficient gives $a_0(L)b_0(L) = 1$ in $\mathbb{K}[L]$. Thus a_0 must be a unit, hence a nonzero constant. Conversely, if $a_0 \in \mathbb{K}^\times$, the triangular reciprocal recurrence in Chapter 4 constructs all $b_n \in \mathbb{K}[L]$. $\square$

The natural field extension is

$$\mathcal{R}_{\mathbb{K}} = \mathbb{K}(L)((q)). \quad (2.6)$$

Every nonzero element of $\mathcal{R}_{\mathbb{K}}$ is invertible because $\mathbb{K}(L)$ is a field. A smaller extension, $\mathbb{K}[L, L^{-1}](q)$, suffices when denominators are monomials in L , but not for a factor such as $L + 1$.

Worked calculation

The series $A = L + q$ has no reciprocal in $\mathcal{P}_{\mathbb{K}}$. In $\mathcal{R}_{\mathbb{K}}$,

$$\frac{1}{L+q} = \frac{1}{L} \frac{1}{1+q/L} = \frac{1}{L} - \frac{q}{L^2} + \frac{q^2}{L^3} - \dots$$

The failure is not caused by the infinite q -expansion; it is already present in the leading coefficient L .

2.5 Affine and monomial prefactors

Many useful expansions do not begin at q -order zero. Write a nonzero series as

$$A = cX^\rho L^\beta (1 + U), \quad U \prec 1, \quad (2.7)$$

whenever such a factorization is available. In the strict ring $\mathcal{P}_{\mathbb{K}}$, $\rho \in \mathbb{Z}$, $\beta \in \mathbb{N}$, and the factor L^β may prevent invertibility. In generalized power-log classes, ρ, β may be real or complex.

The factorization (2.7) separates three tasks:

1. invert or raise the monomial $cX^\rho L^\beta$ explicitly;
2. apply an ordinary binomial, logarithm, or exponential series to $1 + U$;
3. truncate q -adically, using that U has positive valuation.

This is the polynomial-logarithmic analogue of the canonical multiplicative decomposition used in full transseries.

2.6 Degree slopes and finite bookkeeping

The Lambert polynomials satisfy $\deg P_n = n$. More generally, many calculations preserve a linear degree bound

$$\deg a_n \leq \sigma n + C. \quad (2.8)$$

Such a bound is not needed for formal multiplication, but it translates a q -adic remainder into a useful analytic estimate and limits coefficient size.

Proposition 2.3 (Slope under multiplication). *Suppose*

$$\deg a_n \leq A + \alpha n, \quad \deg b_n \leq B + \beta n$$

for $n \geq 0$. If $C = AB = \sum c_n q^n$, then

$$\deg c_n \leq A + B + \max(\alpha, \beta)n.$$

Proof. Each summand $a_i b_{n-i}$ in c_n has degree at most

$$A + B + \alpha i + \beta(n - i) \leq A + B + \max(\alpha, \beta)n.$$

Taking a finite sum cannot increase this bound. □

2.7 Generalized exponent supports

For some problems integer powers of q are too restrictive. Let $S \subset \mathbb{R}$ be a well-ordered set with the property that every real bound contains only finitely many elements of S , or more generally assume the usual Hahn-series support conditions. Then one may consider

$$A(X) = \sum_{\lambda \in S} a_\lambda(L) X^{-\lambda}. \quad (2.9)$$

Multiplication is defined by

$$(AB)_\nu = \sum_{\lambda + \mu = \nu} a_\lambda b_\mu,$$

provided each coefficient receives only finitely many contributions and the sum support remains well ordered. Finite-rank grids

$$S \subset \lambda_0 + \mathbb{N}\omega_1 + \cdots + \mathbb{N}\omega_r, \quad \omega_j > 0,$$

are particularly convenient. The integer q -adic theory developed below then extends with multi-indices or an ordered grid.

A common trap

A formal sum over arbitrary real exponents is not automatically meaningful. Without a support condition, a coefficient of a product may receive infinitely many contributions, or the support may contain an infinite descending chain in the asymptotic order. “Write all powers that seem useful” is not a definition of a transseries algebra.

2.8 Changing to a small variable

Set $x = X^{-1} \rightarrow 0^+$. Then

$$L = \log X = \log(1/x) = -\log x.$$

The core expansion becomes

$$\sum_{n \geq n_0} p_n \left(\log \frac{1}{x} \right) x^n. \quad (2.10)$$

This is the convention common in power-log and Dulac series. Both coordinate systems encode the same algebra, but signs inside logarithmic polynomials change if one uses $\log x$ rather than $\log(1/x)$. We postpone a fuller discussion to [Chapter 9](#).

Chapter summary

The core ring is $\mathcal{P}_{\mathbb{K}} = \mathbb{K}[L]((q))$ with $q = X^{-1}$ and $L = \log X$. Its q -adic valuation makes truncation and convergence purely coefficientwise. It is not a field: a series is a unit only when its first nonzero q -coefficient is a nonzero constant. General division naturally leads to $\mathbb{K}(L)((q))$. Generalized real exponents are possible, but only with support conditions that preserve local finiteness.

Multiplication and differential calculus

3.1 Addition and normalization

Let

$$A = \sum_{n \geq r} a_n(L)q^n, \quad B = \sum_{n \geq s} b_n(L)q^n.$$

After inserting zero coefficients where necessary,

$$A + B = \sum_{n \geq \min(r,s)} (a_n(L) + b_n(L))q^n.$$

One should expand, collect, or otherwise normalize each polynomial coefficient before testing it for zero. This matters because cancellation may raise the q -valuation.

Example 3.1. If

$$A = q^{-1} - L + (L^2 - 1)q, \quad B = -q^{-1} + L + (2 - L^2)q + 3q^2,$$

then

$$A + B = q + 3q^2,$$

so two leading q -blocks and the quadratic logarithmic terms all cancel.

3.2 Cauchy multiplication

Multiplication uses the same convolution as ordinary power series, but the coefficient arithmetic takes place in $\mathbb{K}[L]$.

Proposition 3.1 (Product formula). *For*

$$A = \sum_{i \geq r} a_i(L)q^i, \quad B = \sum_{j \geq s} b_j(L)q^j,$$

the product is

$$AB = \sum_{n \geq r+s} c_n(L)q^n, \quad c_n(L) = \sum_{i+j=n} a_i(L)b_j(L). \quad (3.1)$$

For each fixed n , the sum is finite.

Proof. Because $i \geq r$ and $j \geq s$, the equation $i + j = n$ restricts i to the finite interval $r \leq i \leq n - s$. Distributivity then gives Equation (3.1). The result is again a Laurent series with polynomial coefficients. $\square$

To compute modulo q^N , discard every pair with $i + j \geq N$. No higher coefficient can influence a lower one.

Algorithm 3.1 (Truncated multiplication). Given sparse maps $i \mapsto a_i(L)$ and $j \mapsto b_j(L)$, and a cutoff N :

1. initialize $c_n = 0$ for all required $n < N$;
2. for each stored i and j with $i + j < N$, add $a_i b_j$ to c_{i+j} ;
3. normalize the polynomials and delete zero blocks.

The output is $[AB]_{<N}$.

Worked calculation

Take

$$A = 1 + (L - 1)q + (L^2 + 2)q^2 + O_q(q^3),$$

$$B = 1 - (L + 2)q + (3L - 1)q^2 + O_q(q^3).$$

The coefficient of q is

$$(L - 1) - (L + 2) = -3.$$

The coefficient of q^2 is

$$(L^2 + 2) + (3L - 1) - (L - 1)(L + 2) = 2L + 3.$$

Therefore

$$AB = 1 - 3q + (2L + 3)q^2 + O_q(q^3).$$

The leading logarithmic squares cancel even though neither factor has constant coefficients.

3.3 Powers of a normalized series

If $U \in q\mathbb{K}[L][[q]]$, then $U^m \in q^m\mathbb{K}[L][[q]]$. Consequently, for a fixed cutoff q^N , only $m < N$ can contribute to a series in powers of U . This simple observation makes formal binomial, exponential, and logarithmic series finite at every requested precision.

For an integer $r \geq 0$, $(1 + U)^r$ is an ordinary finite polynomial. For arbitrary $\alpha \in \mathbb{K}$, define

$$(1 + U)^\alpha = \sum_{m \geq 0} \binom{\alpha}{m} U^m, \quad \binom{\alpha}{m} = \frac{\alpha(\alpha - 1) \cdots (\alpha - m + 1)}{m!}. \quad (3.2)$$

Modulo q^N , the sum stops at $m = N - 1$.

Example 3.2 (Square root). Let

$$A = 1 + 2Lq + (L^2 + 1)q^2 + O_q(q^3).$$

Using $(1 + U)^{1/2} = 1 + U/2 - U^2/8 + \dots$,

$$A^{1/2} = 1 + Lq + \frac{1}{2}q^2 + O_q(q^3),$$

because the L^2q^2 contributions cancel.

3.4 Differentiation and the shifted derivative

Let $D = d/dL$. Since $L = \log X$,

$$\frac{dL}{dX} = X^{-1}.$$

For a single block,

$$\frac{d}{dX}(a_n(L)X^{-n}) = (Da_n - na_n)X^{-n-1}. \quad (3.3)$$

It is convenient to define

$$\Delta_n = D - n. \quad (3.4)$$

Then

$$A'(X) = \sum_n \Delta_n a_n(L) X^{-n-1}. \quad (3.5)$$

The index shift is as important as the coefficient derivative.

Derivation of Equation (3.3). The product and chain rules give

$$\frac{d}{dX}(a_n(\log X)X^{-n}) = a'_n(L)X^{-1}X^{-n} - na_n(L)X^{-n-1}.$$

Collecting the common power proves the formula. □

Example 3.3. For

$$A = X - L + \frac{L^2 - 3L}{X},$$

we obtain

$$A' = 1 - \frac{1}{X} + \frac{(2L - 3) - (L^2 - 3L)}{X^2} = 1 - \frac{1}{X} + \frac{-L^2 + 5L - 3}{X^2}.$$

A common trap

If a computer algebra system treats L as a symbol independent of X , it will incorrectly differentiate $a(L)X^{-n}$ as merely $-na(L)X^{-n-1}$. Either substitute $L = \log X$ before differentiating or implement the shifted operator Δ_n explicitly.

3.5 Repeated derivatives

One derivative changes both the power index and the shifted operator. Applying Equation (3.3) repeatedly gives

$$\frac{d^r}{dX^r} (a_n(L)X^{-n}) = \left(\prod_{j=0}^{r-1} (D - n - j) \right) a_n(L) X^{-n-r}. \quad (3.6)$$

The factors commute because each is a polynomial in D . For example,

$$\frac{d^2}{dX^2} (a_n(L)X^{-n}) = (D - n - 1)(D - n) a_n(L) X^{-n-2}.$$

This formula is useful in Taylor expansions during composition.

3.6 Formal antiderivatives

Although integration is not a central operation here, it clarifies why polynomial coefficients are stable under differential calculus. To integrate $p(L)X^{-n}$, seek $Q(L)X^{-n+1}$. Then

$$\frac{d}{dX} (Q(L)X^{-n+1}) = (D - (n - 1))Q(L)X^{-n}.$$

For $n \neq 1$, the operator $D - (n - 1)$ is invertible on $\mathbb{K}[L]$, with finite inverse

$$(D - a)^{-1} p = - \sum_{j=0}^{\deg p} \frac{p^{(j)}}{a^{j+1}}, \quad a \neq 0. \quad (3.7)$$

Indeed, the sum telescopes after applying $D - a$. When $n = 1$,

$$\int p(L)X^{-1} dX = \int p(L) dL,$$

which is again polynomial in L . Thus termwise antiderivatives remain polynomial–logarithmic, up to an arbitrary constant.

3.7 Differential equations for coefficient blocks

Suppose a formal equation produces

$$(D - n)p_n(L) = r_n(L)$$

for a known polynomial r_n . If $n \neq 0$, Equation (3.7) solves uniquely for p_n . If $n = 0$, the equation is $p'_0 = r_0$, which determines p_0 up to a constant. Such resonant constants are often fixed by a normalization condition or by matching another asymptotic region.

This blockwise inversion is the simplest instance of a general principle: formal asymptotic equations become triangular because the dominant monomial separates successive q -orders.

3.8 Products of many factors and Bell polynomials

Repeated multiplication can be organized by partitions. Let

$$U(q) = \sum_{j \geq 1} u_j q^j.$$

The coefficient of q^n in U^r is

$$[q^n]U^r = \sum_{j_1 + \dots + j_r = n} u_{j_1} \cdots u_{j_r}, \quad (3.8)$$

where all $j_i \geq 1$. Equivalently, in terms of the partial exponential Bell polynomial $B_{n,r}$,

$$[q^n]U^r = \frac{r!}{n!} B_{n,r}(1!u_1, 2!u_2, \dots, (n-r+1)!u_{n-r+1}). \quad (3.9)$$

We will use this identity to expand $\log(1+U)$, exponentials, and composition systematically.

3.9 Complexity at fixed truncation

With dense q -coefficients, schoolbook multiplication through order N uses $O(N^2)$ polynomial products. If the logarithmic degree is $O(N)$, naive polynomial multiplication adds another quadratic factor in the worst case. For the moderate orders typical in hand asymptotics, this is acceptable and transparent. At high orders one may use fast convolution in q , fast polynomial multiplication in L , or relaxed power-series algorithms.

The valuation is more important than asymptotic complexity: a correct implementation should never compute terms that cannot affect the requested precision.

Chapter summary

Multiplication is Cauchy convolution in the q -index and ordinary polynomial multiplication in L . Positive q -valuation makes binomial and other power-series substitutions finite at fixed precision. Differentiation is controlled by the shifted operator $\Delta_n = D - n$, because $L = \log X$ is coupled to X . Bell polynomials package the coefficient combinatorics of repeated products.

Reciprocals, division, logarithms, and exponentials

4.1 Multiplicative inversion is a triangular problem

Let

$$A = a_0 + a_1q + a_2q^2 + \cdots, \quad a_0 \in \mathbb{K}^\times, \quad (4.1)$$

and seek

$$A^{\langle -1 \rangle} = B = b_0 + b_1q + b_2q^2 + \cdots$$

such that $AB = 1$. Comparing q-coefficients gives

$$a_0b_0 = 1,$$

and, for $n \geq 1$,

$$a_0b_n + \sum_{k=1}^n a_kb_{n-k} = 0.$$

Therefore

$$b_0 = a_0^{-1}, \quad b_n = -a_0^{-1} \sum_{k=1}^n a_kb_{n-k}. \quad (4.2)$$

Every b_n depends only on earlier b_j , so the calculation is triangular.

Proposition 4.1 (Existence and uniqueness of the reciprocal). *If $a_0 \in \mathbb{K}^\times$, the recurrence (4.2) defines the unique $B \in \mathbb{K}[L][[q]]$ satisfying $AB = 1$.*

Proof. Existence follows by induction: once $b_0, \dots, b_{n-1}$ are known, the coefficient equation at q-order n has the unique solution shown in Equation (4.2). The resulting product has constant coefficient one and every positive q-coefficient zero. If B and $\tilde{B}$ are two reciprocals, then $B = B(AB) = (BA)\tilde{B} = \tilde{B}$. $\square$

A Laurent prefactor causes no difficulty. If

$$A = q^r \tilde{A}, \quad \text{val}_q(\tilde{A}) = 0,$$

then

$$A^{\langle -1 \rangle} = q^{-r} \tilde{A}^{\langle -1 \rangle}.$$

Worked calculation

Let

$$A = 1 + Lq + (L^2 + 1)q^2 + O_q(q^4),$$

with the q -cubic coefficient taken to be zero. The recurrence gives

$$b_0 = 1, \quad b_1 = -L,$$

$$b_2 = -(Lb_1 + (L^2 + 1)b_0) = -1,$$

and

$$b_3 = -(Lb_2 + (L^2 + 1)b_1) = L^3 + 2L.$$

Hence

$$\frac{1}{A} = 1 - Lq - q^2 + (L^3 + 2L)q^3 + O_q(q^4).$$

Direct multiplication is the quickest verification.

4.2 Geometric normalization

When $A = a_0(1 + U)$ with $U \in q\mathbb{K}[L][[q]]$, the recurrence is equivalent to

$$\frac{1}{A} = a_0^{-1} \sum_{m \geq 0} (-U)^m. \quad (4.3)$$

At precision q^N , only $m < N$ contributes. The recurrence is usually better for sequential coefficient generation; the geometric formula is often better for hand calculations and proofs.

4.3 Direct quotient recurrence

To divide C by A , one may compute $CA^{(-1)}$, but a direct recurrence avoids storing the entire reciprocal. Suppose

$$A = \sum_{n \geq 0} a_n q^n, \quad C = \sum_{n \geq 0} c_n q^n, \quad Q = \frac{C}{A} = \sum_{n \geq 0} d_n q^n,$$

with $a_0 \in \mathbb{K}^\times$. From $AQ = C$,

$$d_0 = \frac{c_0}{a_0}, \quad d_n = \frac{1}{a_0} \left(c_n - \sum_{k=1}^n a_k d_{n-k} \right). \quad (4.4)$$

This works without change when the coefficients lie in $\mathbb{K}(L)$ rather than $\mathbb{K}[L]$.

Example 4.1. With

$$C = 1 + (2L + 1)q + L^2 q^2, \quad A = 1 + (L - 1)q + q^2,$$

we find

$$d_0 = 1, \quad d_1 = (2L + 1) - (L - 1) = L + 2,$$

$$d_2 = L^2 - (L - 1)(L + 2) - 1 = -L + 1.$$

Thus

$$\frac{C}{A} = 1 + (L + 2)q + (1 - L)q^2 + O_q(q^3).$$

4.4 What changes when the leading coefficient depends on L

If $a_0(L)$ is a nonconstant polynomial, the same recurrence still works in $\mathbb{K}(L)[[q]]$:

$$b_0 = \frac{1}{a_0(L)}, \quad b_n = -\frac{1}{a_0(L)} \sum_{k=1}^n a_k(L)b_{n-k}(L).$$

The output generally has rational, not polynomial, logarithmic coefficients. For instance,

$$\frac{1}{L + q} = L^{-1} - L^{-2}q + L^{-3}q^2 - \dots.$$

There are three reasonable responses in symbolic work:

1. enlarge the coefficient field to $\mathbb{K}(L)$;
2. factor the logarithmic monomial and work in a class allowing negative powers of L ;
3. keep the expression factored rather than expanding it.

The right choice depends on the intended subsequent operations. Expanding into L^{-1} can be asymptotically useful, but it changes the monomial hierarchy.

4.5 Newton inversion

The recurrence computes one coefficient at a time. Newton's method can double the number of correct coefficients at each step. Given an approximation B to A^{-1} , set

$$B_{\text{new}} = B(2 - AB). \tag{4.5}$$

Proposition 4.2 (Precision doubling for reciprocals). *If*

$$AB = 1 - E, \quad \text{val}_q(E) \geq N,$$

then

$$AB_{\text{new}} = 1 - E^2, \quad \text{val}_q(1 - AB_{\text{new}}) \geq 2N.$$

Proof. Substitute $AB = 1 - E$:

$$AB_{\text{new}} = (AB)(2 - AB) = (1 - E)(1 + E) = 1 - E^2.$$

The valuation is multiplicative, so $\text{val}_q(E^2) \geq 2N$. □

A practical implementation begins with $B_0 = a_0^{-1}$ and truncates after each Newton step to the desired doubled precision. Fast multiplication then makes Newton inversion asymptotically preferable at high order.

4.6 Formal logarithms

If $U \in q\mathbb{K}[L][[q]]$, define

$$\log(1 + U) = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} U^m. \quad (4.6)$$

This lies in $q\mathbb{K}[L][[q]]$. For

$$A = cX^\rho(1 + U), \quad c \neq 0,$$

a chosen logarithm branch gives

$$\log A = \log c + \rho L + \log(1 + U). \quad (4.7)$$

If a factor L^β is present, an additional term $\beta \log L$ appears. That term lies outside the one-log-level coefficient ring and is the first clear signal that iterated logarithms may be required.

Worked calculation

Let

$$A = 1 + Lq + (2 - L)q^2 + O_q(q^3).$$

Then $U = Lq + (2 - L)q^2$, and

$$\log A = U - \frac{U^2}{2} + O_q(q^3) = Lq + \left(2 - L - \frac{L^2}{2}\right)q^2 + O_q(q^3).$$

4.7 Formal exponentials

If $V \in q\mathbb{K}[L][[q]]$, define

$$\exp V = \sum_{m \geq 0} \frac{V^m}{m!}. \quad (4.8)$$

It lies in $1 + q\mathbb{K}[L][[q]]$. At fixed q -precision, only finitely many powers of V contribute. The formal identities

$$\exp(\log(1 + U)) = 1 + U, \quad \log(\exp V) = V$$

follow from the corresponding identities in the complete q -adic algebra.

An arbitrary polynomial in L cannot be exponentiated while staying inside $\mathcal{P}_{\mathbb{K}}$. For example,

$$\exp(L) = X = q^{-1}$$

is allowed after recognizing the monomial, but

$$\exp(L^2) = X^{\log X}$$

is a genuinely new transmonomial. Full transseries handle such exponentials by enlarging the monomial group.

4.8 Logarithmic differentiation

For a unit A ,

$$\frac{A'}{A} = (\log A)'. \quad (4.9)$$

If $A = cX^\rho(1 + U)$, then

$$\frac{A'}{A} = \frac{\rho}{X} + \frac{U'}{1 + U}.$$

The logarithmic derivative often has higher q -valuation than A' itself and is a natural measure of how rapidly a transmonomial changes. In the general theory of composition, comparisons between a perturbation and logarithmic derivatives of monomials distinguish shallow, moderate, and deep substitutions [5].

4.9 General powers

For a normalized unit $A = a_0(1 + U)$ and $\alpha \in \mathbb{K}$, define

$$A^\alpha = a_0^\alpha(1 + U)^\alpha, \quad (4.10)$$

with compatible choices of a_0^α and logarithm branch. Equivalently, $A^\alpha = \exp(\alpha \log A)$. The binomial formula Equation (3.2) gives a direct coefficient algorithm.

For a general prefactor $cX^\rho L^\beta(1 + U)$,

$$A^\alpha = c^\alpha X^{\alpha\rho} L^{\alpha\beta}(1 + U)^\alpha.$$

Closure may require generalized powers of X or L , but the small-unit part remains q -adic.

4.10 Residual-based verification

Every reciprocal or quotient should be checked by multiplication. If B_N is intended as the reciprocal of A through order q^{N-1} , verify

$$AB_N - 1 = O_q(q^N). \quad (4.11)$$

For $Q_N = C/A$, verify

$$AQ_N - C = O_q(q^N).$$

These tests detect off-by-one truncation errors, failures to normalize polynomials, and accidental use of L as an independent variable.

Chapter summary

Reciprocal and quotient coefficients satisfy triangular convolution recurrences. Polynomial coefficients are preserved only when the leading coefficient is a nonzero constant; otherwise the natural coefficient field is $\mathbb{K}(L)$. Newton's update $B \mapsto B(2 - AB)$ squares the residual and doubles q -adic precision. Logarithms and exponentials are defined on small q -adic perturbations, while prefactors determine whether new monomials such as $\log L$ or $\exp(L^2)$ must be admitted.

Composition

5.1 Why composition is the delicate operation

Addition and multiplication keep L fixed. Composition does not. If an outer series is written in a variable Y , then substitution $Y = G(X)$ changes both of its building blocks:

$$Y^{-1} \mapsto G(X)^{-1}, \quad \log Y \mapsto \log G(X).$$

The second change is responsible for derivatives of coefficient polynomials; the first is responsible for shifted derivatives. Moreover, an unrestricted substitution can introduce $\log \log X$, new real powers, or even ill-based supports. We therefore begin with a composition class whose closure can be proved directly.

5.2 Regular inner arguments

Let the outer series be

$$F(Y) = \sum_{n \geq n_0} f_n(\log Y) Y^{-n}, \quad f_n \in \mathbb{K}[L]. \quad (5.1)$$

Assume that the inner argument has the form

$$G(X) = \lambda X(1 + u(X)), \quad \lambda \in \mathbb{K}^\times, \quad u \in q\mathbb{K}[L][[q]]. \quad (5.2)$$

For an analytic interpretation, choose compatible logarithm branches and a region in which $G(X)$ stays in the domain of the outer branch. Formally set

$$\delta = \log(1 + u) \in q\mathbb{K}[L][[q]]. \quad (5.3)$$

Then

$$\log G = L + \log \lambda + \delta, \quad G^{-n} = \lambda^{-n} X^{-n} e^{-n\delta}. \quad (5.4)$$

5.3 The normal-ordered composition formula

Let $D = d/dL$. Because δ itself depends on L , the operators “multiply by δ ” and D do not commute. We therefore introduce an explicit normal-ordering convention.

Definition 5.1 (Normal-ordered shifted exponential). For a polynomial $f(L)$, a formal small series δ , and an integer n , define

$$:e^{\delta(D-n)}: f := \sum_{r \geq 0} \frac{\delta^r}{r!} (D-n)^r f. \quad (5.5)$$

The colons mean that every power of δ is placed to the left and is not differentiated by D .

Even though $(D-n)^r f$ need not vanish for large r , the series is q -adically meaningful because $\text{val}_q(\delta^r) \geq r$.

Theorem 5.1 (Regular composition formula). Under Equations (5.1) and (5.2),

$$(F \circ G)(X) = \sum_{n \geq n_0} \lambda^{-n} X^{-n} :e^{\delta(D-n)}: f_n(L + \log \lambda). \quad (5.6)$$

At every finite q -precision, the right side requires only finitely many terms.

Proof. For one outer block,

$$f_n(\log G) G^{-n} = \lambda^{-n} X^{-n} f_n(L + \log \lambda + \delta) e^{-n\delta}.$$

Taylor-expand the polynomial coefficient:

$$f_n(L + \log \lambda + \delta) = \sum_{j \geq 0} \frac{\delta^j}{j!} D^j f_n(L + \log \lambda).$$

Multiplying by $e^{-n\delta} = \sum_{k \geq 0} (-n)^k \delta^k / k!$ and collecting the total power $r = j + k$ gives

$$\sum_{r \geq 0} \frac{\delta^r}{r!} (D-n)^r f_n(L + \log \lambda),$$

by the binomial theorem. This is precisely the normal-ordered expression in Equation (5.6). To compute modulo q^N , only outer indices $n < N$ and powers $r < N - n$ can contribute, up to the finite negative part of the Laurent series. $\square$

A common trap

Without the normal-ordering colons, the expression $\exp(\delta(D-n))$ is ambiguous and generally wrong. If interpreted as an ordinary differential-operator exponential, repeated application of δD differentiates δ . Taylor substitution requires D to act only on the outer coefficient polynomial.

5.4 Coefficient extraction with Bell polynomials

Write

$$\delta(q) = \sum_{j \geq 1} \delta_j(L) q^j.$$

By Equation (3.9),

$$[q^m]\delta^r = \frac{r!}{m!} B_{m,r}(1!\delta_1, 2!\delta_2, \dots).$$

Consequently, the coefficient of q -order N in Equation (5.6) is

$$[q^N](F \circ G) = \sum_{n=n_0}^N \lambda^{-n} \frac{1}{(N-n)!} \sum_{r=0}^{N-n} B_{N-n,r}(1!\delta_1, 2!\delta_2, \dots) \times (D-n)^r f_n(L + \log \lambda), \quad (5.7)$$

with $B_{0,0} = 1$ and $B_{m,0} = 0$ for $m > 0$. This is a closed, finite formula for every coefficient.

Algorithm 5.1 (Regular composition to q -order N).

1. Normalize the inner series as $G = \lambda X(1 + u)$.
2. Compute $\delta = [\log(1 + u)]_{<N}$.
3. For every outer block $f_n(\log Y)Y^{-n}$ that can contribute, compute the shifted derivatives $(D-n)^r f_n(L + \log \lambda)$.
4. Multiply them by $\delta^r/r!$, the factor $\lambda^{-n}q^n$, and truncate after every multiplication.
5. Sum and normalize the coefficient polynomials.

Bell polynomials may replace repeated powers of δ when many coefficients are required.

5.5 A first worked composition

Take

$$F(Y) = \log Y + \frac{(\log Y)^2}{Y}, \quad G(X) = 2X \left(1 + \frac{L}{X}\right).$$

Here $\lambda = 2$, $u = Lq$, and

$$\delta = \log(1 + Lq) = Lq - \frac{L^2}{2}q^2 + O_q(q^3).$$

Directly,

$$\log G = L + \log 2 + \delta.$$

Also,

$$G^{-1} = \frac{q}{2}e^{-\delta} = \frac{q}{2} \left(1 - Lq + \frac{3L^2}{2}q^2 + O_q(q^3)\right).$$

Therefore, through q -order two,

$$\begin{aligned} F(G) &= L + \log 2 + Lq - \frac{L^2}{2}q^2 \\ &\quad + \frac{q}{2}(L + \log 2 + \delta)^2 e^{-\delta} + O_q(q^3) \\ &= L + \log 2 + \left[L + \frac{1}{2}(L + \log 2)^2\right]q \\ &\quad + \left[-\frac{L^2}{2} + L(L + \log 2) - \frac{L}{2}(L + \log 2)^2\right]q^2 + O_q(q^3). \end{aligned}$$

The shifted-operator formula produces the same result without separately expanding the logarithmic coefficient and the reciprocal power.

5.6 Near-identity inner arguments

The most important case for series reversal is

$$G(X) = X + A(L) + \frac{B(L)}{X} + \frac{C(L)}{X^2} + \frac{D(L)}{X^3} + O_q(q^4). \quad (5.8)$$

Factoring out X gives

$$G = X(1 + Aq + Bq^2 + Cq^3 + Dq^4 + \dots).$$

Thus

$$\begin{aligned} \delta = \log(G/X) &= Aq + \left(B - \frac{A^2}{2}\right)q^2 + \left(C - AB + \frac{A^3}{3}\right)q^3 \\ &\quad + \left(D - AC - \frac{B^2}{2} + A^2B - \frac{A^4}{4}\right)q^4 + O_q(q^5). \end{aligned} \quad (5.9)$$

Every capital letter here is a polynomial in L , and the products are ordinary products at the same value of L .

5.7 Universal near-identity composition coefficients

Let the outer series be

$$F(Y) = Y + a(\log Y) + \frac{b(\log Y)}{Y} + \frac{c(\log Y)}{Y^2} + \frac{d(\log Y)}{Y^3} + O(Y^{-4}), \quad (5.10)$$

and let G be as in Equation (5.8). Write

$$F(G(X)) = X + h_0(L) + \frac{h_1(L)}{X} + \frac{h_2(L)}{X^2} + \frac{h_3(L)}{X^3} + O_q(q^4).$$

Primes below mean derivatives with respect to L , and all uncapitalized functions are evaluated at L .

Theorem 5.2 (Composition through fourth block). *The coefficients are*

$$h_0 = A + a, \quad (5.11)$$

$$h_1 = Aa' + B + b, \quad (5.12)$$

$$h_2 = -\frac{1}{2}A^2a' + \frac{1}{2}A^2a'' - Ab + Ab' + Ba' + C + c, \quad (5.13)$$

$$\begin{aligned} h_3 = & \frac{1}{3}A^3a' - \frac{1}{2}A^3a'' + \frac{1}{6}A^3a''' + A^2b - \frac{3}{2}A^2b' + \frac{1}{2}A^2b'' \\ & - ABA' + ABA'' - 2Ac + Ac' - Bb + Bb' + Ca' + D + d. \end{aligned} \quad (5.14)$$

Proof. Use $G = X(1 + u)$ with $u = Aq + Bq^2 + Cq^3 + Dq^4 + \cdots$. Expand $\delta = \log(1 + u)$ by Equation (5.9), then Taylor-expand $a(L + \delta)$, $b(L + \delta)$, $\dots$, and multiply by $G^{-j} = q^j(1 + u)^{-j}$. At q -order three, one needs derivatives through a''' , b'' , and c' . Collecting powers gives Equations (5.11) to (5.14). The calculation is finite and is also an immediate specialization of Equation (5.6). $\square$

These formulas display an important asymmetry: derivatives fall on the outer coefficient functions a, b, c, d , not on the inner functions A, B, C, D . This is because the outer logarithmic argument is Taylor-expanded around L ; the inner coefficients merely supply the increment.

One step toward Lambert W

Let

$$F(Y) = Y + \log Y, \quad G(X) = X - L + \frac{L}{X}.$$

In theorem 5.2, the outer coefficients are $a(L) = L$ and $b = c = d = 0$, while $A = -L$, $B = L$, and $C = D = 0$. Hence

$$h_0 = -L + L = 0, \quad h_1 = (-L)(1) + L = 0,$$

and

$$h_2 = -\frac{L^2}{2} + L.$$

Thus

$$F(G) = X + \left(L - \frac{L^2}{2}\right)X^{-2} + \mathcal{O}(X^{-3}).$$

Adding

$$C(L) = \frac{L^2}{2} - L = \frac{L(L - 2)}{2}$$

to the coefficient of X^{-2} in G cancels this residual. This is the second Lambert polynomial.

5.8 Formal Taylor composition

For a near-identity substitution $G = X + H$ with $H/X \in q\mathbb{K}[L][[q]]$, one may also write

$$F(X + H) = \sum_{r \geq 0} \frac{F^{(r)}(X)}{r!} H^r. \quad (5.15)$$

This is q -adically meaningful for polynomial–logarithmic F : each derivative of a block raises its q -order by one, while H has q -valuation at least zero, so the r -th Taylor term gains at least r q -orders relative to the same outer block.

Proposition 5.1 (Taylor formula in the near-identity class). *If $F \in \mathcal{P}_{\mathbb{K}}$ and $H/X \in q\mathbb{K}[L][[q]]$, then Equation (5.15) agrees coefficientwise with direct substitution $F \circ (X + H)$.*

Proof. For a single block $f_n(\log X)X^{-n}$, the claim is the ordinary formal Taylor identity after replacing X by an indeterminate and remembering the chain rule Equation (3.6). Modulo q^N ,

only finitely many outer blocks and derivative orders contribute, so the identity extends by finite summation. Passing to all N proves coefficientwise equality. $\square$

The Taylor form is conceptually simple and leads directly to Newton reversion. The normal-ordered form is often more efficient because it packages the effect of the logarithmic change and the reciprocal power in one shifted operator.

5.9 Associativity, identity, and the chain rule

Within the regular or near-identity composition classes above,

$$(F \circ G) \circ H = F \circ (G \circ H), \quad F \circ X = X \circ F = F.$$

These statements may be checked at every finite q -precision, where all substitutions reduce to ordinary finite algebra. Differentiation satisfies

$$(F \circ G)' = (F' \circ G)G'. \quad (5.16)$$

Again, the proof holds blockwise and then extends by q -adic completeness. These properties turn suitable near-identity series into a group under composition once compositional inverses are constructed.

5.10 Scaling and power-leading inner arguments

The same method works for

$$G(X) = \lambda X^p (1 + u), \quad p > 0. \quad (5.17)$$

Now

$$\log G = pL + \log \lambda + \delta, \quad G^{-n} = \lambda^{-n} X^{-pn} e^{-n\delta}.$$

Thus

$$F \circ G = \sum_n \lambda^{-n} X^{-pn} : e^{\delta(D-n)} : f_n(pL + \log \lambda), \quad (5.18)$$

where D differentiates the displayed argument. If p is not an integer, generalized q -exponents are required, but no new logarithmic level appears.

5.11 The first closure failure: a logarithmic prefactor

Suppose instead that

$$G(X) = \lambda X^p L^\beta (1 + u). \quad (5.19)$$

Then

$$\log G = pL + \beta \log L + \log \lambda + \delta. \quad (5.20)$$

If any outer coefficient depends nontrivially on $\log Y$, substitution creates $\log L = \log \log X$. The one-level ring $\mathbb{K}[L](q)$ is therefore not closed under this composition unless $\beta = 0$ or the outer logarithmic dependence happens to cancel.

Example 5.1. Take $F(Y) = \log Y$ and $G(X) = XL$. Then

$$F(G) = \log X + \log \log X = L + \log L,$$

which cannot be expressed with polynomial coefficients in L alone.

The correct remedy is not to suppress the new term but to enlarge the coefficient algebra to include $L_2 = \log L$, and, if necessary, further iterates $L_3 = \log L_2$, and so on. This is one route from elementary power-log series to logarithmic transseries.

5.12 Why arbitrary infinite substitution can fail

Even if each individual monomial can be substituted, an infinite sum may fail to define a valid series. Two pathologies are typical:

1. infinitely many distinct outer monomials may collapse onto the same output monomial, producing an undefined infinite coefficient sum;
2. the substituted support may cease to be well ordered in the asymptotic direction.

The q -adic class with $G \sim \lambda X$ avoids these problems because, below any cutoff, only finitely many outer q -blocks and Taylor orders contribute. General transseries composition requires more sophisticated support theorems; Edgar's treatment gives a detailed account [5].

5.13 A composition checklist

Before composing, answer the following questions.

1. Is the inner argument asymptotically large in the intended analytic region?
2. Can it be factored as $\lambda X^p(1+u)$ with $u \prec 1$?
3. Does $\log G$ introduce $\log L$ or a further logarithmic level?
4. Do the output q -exponents remain in the chosen support grid?
5. At the requested precision, which outer blocks and Taylor orders can actually contribute?
6. Has the result been checked by an independent expansion or by a functional residual?

Chapter summary

For $G = \lambda X(1+u)$, composition is governed by

$$F \circ G = \sum_n \lambda^{-n} X^{-n} : e^{\delta(D-n)} : f_n(L + \log \lambda), \quad \delta = \log(1+u).$$

The normal ordering is essential because δ depends on L . Bell polynomials give a finite coefficient formula. Near the identity, explicit universal coefficients reveal how derivatives of the outer logarithmic polynomials enter. A leading factor L^β in the inner argument creates $\log L$ and forces a larger iterated-log class.

Series reversal and compositional inverses

6.1 Three meanings of “inverse”

The word *inverse* is used for several different operations:

$$-A \quad (\text{additive inverse}), \quad A^{\langle -1 \rangle} = \frac{1}{A} \quad (\text{multiplicative inverse}),$$

$$F^{[-1]} \quad (\text{compositional inverse}).$$

This chapter concerns the last operation. It is also called *series reversion* or *series reversal*: given F , find G such that

$$F \circ G = X, \quad G \circ F = X. \quad (6.1)$$

It does not mean writing the coefficients in reverse order.

6.2 The near-identity composition group

Define

$$\mathcal{G}_{\mathbb{K}} = \left\{ X + \sum_{n \geq 0} f_n(L) X^{-n} : f_n \in \mathbb{K}[L] \right\}. \quad (6.2)$$

Every $F \in \mathcal{G}_{\mathbb{K}}$ satisfies $F/X = 1 + o(1)$, so the composition formulas of Chapter 5 apply. The set is closed under composition, and X is the identity element.

The key triangular fact is that a new inner coefficient first appears unchanged at its own q -order.

Lemma 6.1 (Leading effect of a correction). *Let $F \in \mathcal{G}_{\mathbb{K}}$, let $G \in \mathcal{G}_{\mathbb{K}}$, and let $H = C(L)X^{-N}$ with $N \geq 0$. Then*

$$F(G + H) = F(G) + C(L)X^{-N} + O_q(q^{N+1}). \quad (6.3)$$

Proof. The formal Taylor formula gives

$$F(G + H) = F(G) + (F' \circ G)H + \frac{1}{2}(F'' \circ G)H^2 + \cdots.$$

For $F \in \mathcal{G}_{\mathbb{K}}$, $F' = 1 + O_q(q)$, while $F'' = O_q(q^2)$. Thus the linear term is $H + O_q(q^{N+1})$, and all higher Taylor terms have q -order strictly greater than N . $\square$

Theorem 6.1 (Existence and uniqueness of near-identity reversal). *Every $F \in \mathcal{G}_{\mathbb{K}}$ has a unique compositional inverse $F^{[-1]} \in \mathcal{G}_{\mathbb{K}}$.*

Proof. Construct

$$G_N = X + A_0(L) + A_1(L)q + \cdots + A_N(L)q^N$$

inductively so that

$$F \circ G_N = X + O_q(q^{N+1}).$$

At order zero, choose A_0 to cancel the q -zero residual. Suppose the coefficients through A_{N-1} have been chosen and

$$F \circ G_{N-1} = X + r_N(L)q^N + O_q(q^{N+1}).$$

By lemma 6.1, replacing G_{N-1} by $G_N = G_{N-1} - r_N q^N$ cancels the residual at order N without altering lower orders. Q -adic completeness produces a formal right inverse G .

For uniqueness, if G and $\tilde{G}$ first differ by $C(L)q^N$, then lemma 6.1 shows that $F \circ G$ and $F \circ \tilde{G}$ first differ by the same nonzero block, a contradiction. Finally,

$$F \circ (G \circ F) = (F \circ G) \circ F = F = F \circ X.$$

The same leading-difference argument shows that left composition by F is injective, hence $G \circ F = X$. Therefore the right inverse is two-sided. $\square$

6.3 Residual lifting: one block at a time

The proof gives a particularly simple algorithm.

Algorithm 6.1 (Triangular series reversal). To compute $G = F^{[-1]}$ through q -order N :

1. Start with $G_{-1} = X$.
2. For $n = 0, 1, \dots, N$, compute the residual

$$R_n = F \circ G_{n-1} - X \pmod{q^{n+1}}.$$

3. Extract its q -order n coefficient $r_n(L) = [q^n]R_n$.
4. Set

$$G_n = G_{n-1} - r_n(L)q^n.$$

Then $F \circ G_N = X + O_q(q^{N+1})$.

This algorithm is ideal for derivations and verification because it exposes the residual that determines each new coefficient. It uses repeated composition, so a high-order implementation should switch to Newton iteration.

6.4 Universal inverse coefficients

Let

$$F(X) = X + a(L) + \frac{b(L)}{X} + \frac{c(L)}{X^2} + \frac{d(L)}{X^3} + O_q(q^4), \quad (6.4)$$

and write

$$F^{[-1]}(X) = X + A(L) + \frac{B(L)}{X} + \frac{C(L)}{X^2} + \frac{D(L)}{X^3} + O_q(q^4). \quad (6.5)$$

Theorem 6.2 (Reversal through fourth block). *The coefficients in Equation (6.5) are*

$$A = -a, \quad (6.6)$$

$$B = aa' - b, \quad (6.7)$$

$$C = \frac{1}{2}a^2a' - \frac{1}{2}a^2a'' - a(a')^2 - ab + ab' + a'b - c, \quad (6.8)$$

$$\begin{aligned} D = & \frac{1}{3}a^3a' - \frac{1}{2}a^3a'' + \frac{1}{6}a^3a''' - \frac{3}{2}a^2(a')^2 + \frac{3}{2}a^2a'a'' \\ & - a^2b + \frac{3}{2}a^2b' - \frac{1}{2}a^2b'' + a(a')^3 + 3aa'b - 2aa'b' - aa''b \\ & - 2ac + ac' - (a')^2b + a'c - b^2 + bb' - d. \end{aligned} \quad (6.9)$$

Proof. Substitute Equations (6.4) and (6.5) into the universal composition coefficients Equations (5.11) to (5.14). Set $h_0 = h_1 = h_2 = h_3 = 0$ and solve successively for A, B, C, D . The system is triangular: A occurs only in the first equation, B first occurs linearly in the second, and so on. Straight simplification gives the displayed formulas. $\square$

Checkpoint

The derivatives in Equations (6.6) to (6.9) are derivatives of the *original* coefficient functions a, b, c, d with respect to L . No derivative of an unknown inverse coefficient is needed. This makes the formulas directly evaluable.

6.5 A parameterized example

Consider

$$F(X) = X + \alpha L + \frac{\beta}{X} + O_q(q^2), \quad (6.10)$$

where $\alpha, \beta \in \mathbb{K}$. Here $a = \alpha L$, $a' = \alpha$, and $b = \beta$. The inverse begins

$$\begin{aligned} F^{[-1]}(X) = & X - \alpha L + \frac{\alpha^2 L - \beta}{X} \\ & + \frac{\alpha^3(\frac{1}{2}L^2 - L) - \alpha\beta L + \alpha\beta}{X^2} + O_q(q^3). \end{aligned} \quad (6.11)$$

A direct substitution into Equation (6.10) verifies that the residual is $O_q(q^3)$.

6.6 The Lagrange–Bürmann reversion formula at infinity

There is also a closed differential expression for all coefficients. Write

$$F(X) = X + h(X), \quad h(X) = o(X). \quad (6.12)$$

For our core class, $h \in \mathbb{K}[L][[q]]$: it begins at q -order zero, but is small relative to $X = q^{-1}$.

Theorem 6.3 (Differential reversion formula). *For $F = X + h \in \mathcal{G}_{\mathbb{K}}$,*

$$F^{[-1]}(X) = X + \sum_{m \geq 1} \frac{(-1)^m}{m!} \left(\frac{d}{dX} \right)^{m-1} (h(X)^m). \quad (6.13)$$

The m -th summand has q -valuation at least $m - 1$, so only finitely many summands contribute at any fixed precision.

Proof. Introduce a formal parameter t and consider $F_t(Y) = Y + th(Y)$. The classical Lagrange–Bürmann formula for the inverse of a perturbation of the identity gives

$$F_t^{[-1]}(X) = X + \sum_{m \geq 1} \frac{(-t)^m}{m!} \left(\frac{d}{dX} \right)^{m-1} h(X)^m.$$

For polynomial–logarithmic h , each derivative raises q -order by at least one, so the coefficient of any fixed q -power is a finite polynomial in t . Setting $t = 1$ is therefore formally legitimate and yields Equation (6.13). The result is the unique inverse from [theorem 6.1](#). $\square$

Remark 6.1. Formula (6.13) is often the fastest route to a symbolic closed form for a family of inverse coefficients. The residual-lifting algorithm is usually the clearest route to a proof by induction. Newton iteration, developed next, is usually the fastest route to many coefficients.

6.7 Newton reversion

Let G approximate $F^{[-1]}$. Newton’s correction for the equation $F(G) = X$ is

$$G_{\text{new}} = G - \frac{F \circ G - X}{F' \circ G}. \quad (6.14)$$

The denominator is a unit because $F' = 1 + O_q(q)$.

Theorem 6.4 (Precision doubling for reversion). *Suppose*

$$E = F \circ G - X, \quad \text{val}_q(E) \geq N \geq 1.$$

Then the Newton update Equation (6.14) satisfies

$$\text{val}_q(F \circ G_{\text{new}} - X) \geq 2N. \quad (6.15)$$

Proof. Set

$$\Delta = -\frac{E}{F' \circ G}.$$

Since the denominator is a unit, $\text{val}_q(\Delta) \geq N$. Taylor expansion around G gives

$$F(G + \Delta) - X = E + (F' \circ G)\Delta + \sum_{r \geq 2} \frac{F^{(r)} \circ G}{r!} \Delta^r.$$

The first two terms cancel. Every remaining term contains Δ^r with r at least two, hence has q -valuation at least $2N$; in fact the derivatives of a near-identity series improve the bound further. This proves Equation (6.15). $\square$

A practical schedule is:

1. determine the q -zero correction by residual lifting;
2. compute one more block if necessary so that the residual has positive q -valuation;
3. apply Newton updates at precisions $2, 4, 8, \dots$, truncating all intermediate operations.

6.8 Left and right residuals

To verify a computed inverse G_N , check both

$$F \circ G_N - X = O_q(q^{N+1}), \quad G_N \circ F - X = O_q(q^{N+1}). \quad (6.16)$$

In exact formal algebra, one side and uniqueness imply the other. In software, the two computations exercise different code paths and are therefore valuable independent tests.

6.9 Reversion away from the identity

The near-identity group is not the only invertible asymptotic class. The first step in a more general problem is to invert the dominant monomial.

6.9.1 Algebraic leading term

If

$$F(Y) \sim cY^p, \quad c \neq 0, \quad p > 0,$$

then a branch of the inverse begins

$$Y \sim c^{-1/p} X^{1/p}.$$

Set $Y = c^{-1/p} X^{1/p} (1 + U)$, substitute, and solve for the small series U . Fractional q -powers generally appear.

6.9.2 Power times logarithm

If

$$F(Y) \sim cY^p(\log Y)^\beta, \quad (6.17)$$

then a first balance gives

$$Y \sim \left(\frac{p^\beta}{c}\right)^{1/p} X^{1/p}(\log X)^{-\beta/p}. \quad (6.18)$$

Indeed, $\log Y \sim p^{-1} \log X$. The next correction generally contains $\log \log X / \log X$. Thus the natural inverse class is larger than $\mathbb{K}[L]((X^{-1}))$.

6.9.3 Branch information

Formal reversal determines a family of possible expansions once roots and logarithms are chosen. Analytic inversion requires a domain on which F is one-to-one and its derivative does not vanish. Different choices of roots or logarithm branches can produce different formal inverse sectors.

6.10 A reversion checklist

1. Distinguish compositional inversion from reciprocal formation.
2. Normalize the dominant term before assuming a near-identity form.
3. Choose a support class large enough for fractional powers or iterated logarithms created by the leading inversion.
4. Use residual lifting for transparent coefficient derivation.
5. Use Newton reversion for high order.
6. Verify at least one composition residual, preferably both.
7. Keep analytic branch choices separate from formal coefficient algebra.

Chapter summary

Near-identity polynomial–logarithmic series form a composition group. Their inverse exists uniquely because a correction $C(L)X^{-N}$ first changes the composition residual by exactly the same block. Universal inverse formulas through q-order three follow from the composition formulas. The Lagrange–Bürmann identity

$$(X + h)^{[-1]} = X + \sum_{m \geq 1} \frac{(-1)^m}{m!} \left(\frac{d}{dX} \right)^{m-1} h^m$$

provides a closed all-orders expression, while Newton’s method doubles formal precision. Non-tangent leading terms must first be inverted at the monomial level and may force fractional powers or iterated logarithms.

Lambert W as the central example

7.1 Definition and branches

Lambert's W function is the multivalued inverse of

$$w \mapsto we^w.$$

Thus every branch satisfies

$$W_k(z)e^{W_k(z)} = z, \quad k \in \mathbb{Z}. \quad (7.1)$$

The principal branch is W_0 . On the real interval $(-e^{-1}, 0)$, there is a second real branch W_{-1} . The branch geometry, singularity at $-e^{-1}$, numerical evaluation, and standard asymptotic formulas are developed in Corless et al. and in the NIST Digital Library of Mathematical Functions [2, 3].

For a chosen branch and a large complex argument, introduce

$$X = \xi_k := \text{Log } z + 2\pi ik, \quad L = \log X, \quad (7.2)$$

with compatible logarithm branches. Taking a logarithm of Equation (7.1) gives the asymptotic inverse equation

$$W + \log W = X. \quad (7.3)$$

The same equation defines the Wright omega function once branch unwinding is handled globally. Formally, it is the inverse of $Y \mapsto Y + \log Y$.

7.2 The ansatz and its fixed-point equation

Since $W \sim X$, the first correction in Equation (7.3) is $-\log X = -L$. Write

$$W = X - L + R, \quad R = \sum_{n \geq 1} P_n(L)X^{-n}. \quad (7.4)$$

Then

$$\log W = L + \log \left(1 - \frac{L - R}{X} \right).$$

Substitution into Equation (7.3) yields

$$R = -\log(1 - q(L - R)), \quad q = X^{-1}. \quad (7.5)$$

This single equation determines all Lambert polynomials P_n .

Proposition 7.1 (Triangular Lambert recurrence). *Set $R_{n-1} = \sum_{j=1}^{n-1} P_j(L)q^j$. Then*

$$P_n(L) = [q^n] \left[-\log(1 - q(L - R_{n-1})) \right]. \quad (7.6)$$

The right side involves only $P_1, \dots, P_{n-1}$.

Proof. In $q(L - R)$, the occurrence of $P_n q^n$ is shifted to q -order $n + 1$. Therefore the coefficient of q^n on the right of Equation (7.5) depends only on earlier polynomials. Matching that coefficient with the left side gives Equation (7.6). $\square$

The first four Lambert polynomials

From Equation (7.5),

$$R = q(L - R) + \frac{q^2}{2}(L - R)^2 + \frac{q^3}{3}(L - R)^3 + \dots$$

At q -order one,

$$P_1 = L.$$

At q -order two,

$$P_2 = -P_1 + \frac{L^2}{2} = \frac{L(L - 2)}{2}.$$

At q -order three,

$$P_3 = -P_2 - LP_1 + \frac{L^3}{3} = \frac{L(2L^2 - 9L + 6)}{6}.$$

A further collection gives

$$P_4 = \frac{L(3L^3 - 22L^2 + 36L - 12)}{12}.$$

7.3 Derivation by series reversal

Equation (7.3) asks for the inverse of

$$F(Y) = Y + \log Y.$$

Apply the differential reversion formula Equation (6.13) with $h(X) = \log X = L$:

$$W(e^X) = X + \sum_{m \geq 1} \frac{(-1)^m}{m!} \left(\frac{d}{dX} \right)^{m-1} L^m. \quad (7.7)$$

The term $m = 1$ is $-L$. For $n = m - 1 \geq 1$, comparison with Equation (7.4) gives the compact operator formula

$$P_n(L) = \frac{(-1)^{n+1}}{(n+1)!} \prod_{j=0}^{n-1} (D - j) L^{n+1}, \quad D = \frac{d}{dL}. \quad (7.8)$$

This formula computes P_n independently of all earlier polynomials.

7.4 Stirling-cycle-number formula

Let $\left[\begin{smallmatrix} n \\ j \end{smallmatrix} \right]$ denote the unsigned Stirling number of the first kind, the number of permutations of n elements with exactly j cycles. The signed Stirling numbers $s(n, j)$ satisfy

$$\prod_{r=0}^{n-1} (t - r) = \sum_{j=0}^n s(n, j) t^j, \quad s(n, j) = (-1)^{n-j} \left[\begin{smallmatrix} n \\ j \end{smallmatrix} \right].$$

Expanding the operator product in Equation (7.8) yields the standard closed formula.

Theorem 7.1 (Lambert polynomial formula). *For $n \geq 1$,*

$$P_n(L) = \sum_{m=1}^n (-1)^{n+m} \left[\begin{smallmatrix} n \\ n-m+1 \end{smallmatrix} \right] \frac{L^m}{m!}. \quad (7.9)$$

Consequently,

$$W_k(z) \sim X - L + \sum_{n \geq 1} \frac{1}{X^n} \sum_{m=1}^n (-1)^{n+m} \left[\begin{smallmatrix} n \\ n-m+1 \end{smallmatrix} \right] \frac{L^m}{m!}, \quad (7.10)$$

where $X = \text{Log } z + 2\pi i k$ and $L = \log X$.

Proof. From Equation (7.8),

$$P_n(L) = \frac{(-1)^{n+1}}{(n+1)!} \sum_{j=0}^n s(n, j) D^j L^{n+1}.$$

The $j = 0$ term vanishes because the falling-factorial operator begins with D , and

$$D^j L^{n+1} = \frac{(n+1)!}{(n+1-j)!} L^{n+1-j}.$$

Put $m = n + 1 - j$. Then $j = n - m + 1$, and the sign becomes

$$(-1)^{n+1} (-1)^{n-j} = (-1)^{n+m}.$$

This gives Equation (7.9). Substitution into the ansatz proves Equation (7.10). $\square$

This formula was historically connected with the inversion of $y^\alpha e^y$ and associated Stirling numbers [6, 2]. It is also the form recorded in standard references such as the DLMF [3].

7.5 Structural properties of the Lambert polynomials

Several facts are immediate from Equation (7.9).

Proposition 7.2. *For $n \geq 1$:*

1. P_n has degree n and is divisible by L ;
2. its leading term is L^n/n ;
3. its linear term is $(-1)^{n+1}L$;

4. the nonzero coefficients alternate in sign as the power of L increases;
5. $n!P_n(L) \in \mathbb{Z}[L]$.

Proof. The term with $m = n$ uses $\begin{bmatrix} n \\ 1 \end{bmatrix} = (n-1)!$ and gives L^n/n . The term with $m = 1$ uses $\begin{bmatrix} n \\ n \end{bmatrix} = 1$. Every summand has $m \geq 1$, proving divisibility by L , and its sign is $(-1)^{n+m}$. Finally, $n!/m!$ is an integer for $m \leq n$, so multiplying Equation (7.9) by $n!$ clears all denominators. $\square$

7.6 The expansion through tenth order

The first ten coefficient polynomials are listed here for reference:

$$\begin{aligned}
 P_1 &= L, \\
 P_2 &= \frac{L(L-2)}{2}, \\
 P_3 &= \frac{L(2L^2-9L+6)}{6}, \\
 P_4 &= \frac{L(3L^3-22L^2+36L-12)}{12}, \\
 P_5 &= \frac{L(12L^4-125L^3+350L^2-300L+60)}{60}, \\
 P_6 &= \frac{L}{120}(20L^5-274L^4+1125L^3-1700L^2+900L-120), \\
 P_7 &= \frac{L}{840}(120L^6-2058L^5+11368L^4-25725L^3 \\
 &\quad + 24500L^2-8820L+840), \\
 P_8 &= \frac{L}{2520}(315L^7-6534L^6+45962L^5-142149L^4 \\
 &\quad + 205800L^3-135240L^2+35280L-2520), \\
 P_9 &= \frac{L}{2520}(280L^8-6849L^7+59062L^6-235494L^5 \\
 &\quad + 471429L^4-476280L^3+229320L^2 \\
 &\quad - 45360L+2520), \\
 P_{10} &= \frac{L}{10080}(1008L^9-28516L^8+293175L^7-1447360L^6 \\
 &\quad + 3770550L^5-5314932L^4+3969000L^3 \\
 &\quad - 1461600L^2+226800L-10080).
 \end{aligned}$$

7.7 Formal asymptotics versus actual convergence

Most asymptotic series in applications are divergent, so one should not assume convergence merely from a formal derivation. The Lambert expansion is unusually well behaved. The DLMF records absolute convergence of the standard double series for sufficiently large $|z|$, and convergence on the principal real branch for $z \geq e$ [3]. The formal derivation remains useful

outside a proved convergence region because it determines asymptotic coefficients and branch structure.

Even for a convergent series, the absolute error of successive truncations need not decrease monotonically at every small argument. Cancellation can make one partial sum better than the next.

Table 7.1: Absolute error of the truncation $W_N = X - L + \sum_{n=1}^N P_n(L)X^{-n}$ on the principal real branch.

N	$z = 10$	$z = 10^3$	$z = 10^{20}$
0	2.770×10^{-1}	2.745×10^{-1}	8.482×10^{-2}
2	6.468×10^{-3}	3.923×10^{-3}	3.530×10^{-6}
4	3.255×10^{-3}	1.196×10^{-4}	1.247×10^{-7}
6	7.427×10^{-4}	4.019×10^{-6}	1.186×10^{-10}
8	1.115×10^{-4}	1.183×10^{-7}	6.340×10^{-13}
10	2.176×10^{-6}	1.584×10^{-9}	1.728×10^{-15}

7.8 Residuals and a posteriori correction

For a truncation W_N , define the logarithmic residual

$$\mathcal{R}_N = W_N + \log W_N - X. \quad (7.11)$$

Formal construction gives $\mathcal{R}_N = O_q(q^{N+1})$. Since the derivative of $w + \log w - X$ is $1 + 1/w$, one Newton correction is

$$W_N^{\text{new}} = W_N - \frac{W_N + \log W_N - X}{1 + 1/W_N}. \quad (7.12)$$

As a formal series, this doubles the q -adic precision. Numerically, the residual also provides an inexpensive a posteriori diagnostic. On a domain where $|1 + 1/w|$ is bounded away from zero, the true error is approximately $-\mathcal{R}_N/(1 + 1/W_N)$.

One may instead use the original residual $W_N e^{W_N} - z$, but Equation (7.11) is usually better scaled for large z .

7.9 The real branch W_{-1} near zero

Let $x \rightarrow 0^-$ and define

$$T = \log \frac{1}{-x}, \quad L = \log T. \quad (7.13)$$

Writing $W_{-1}(x) = -u$ transforms $We^W = x$ into

$$u - \log u = T.$$

Equivalently, use the general branch formula with the real asymptotic coordinates. The result is

$$W_{-1}(x) \sim -T - L + \sum_{n \geq 1} \frac{(-1)^n P_n(L)}{T^n}. \quad (7.14)$$

Thus

$$W_{-1}(x) = -T - L - \frac{L}{T} + \frac{L(L-2)}{2T^2} - \frac{L(2L^2 - 9L + 6)}{6T^3} + \frac{L(3L^3 - 22L^2 + 36L - 12)}{12T^4} + \dots \quad (7.15)$$

The alternating powers of T^{-1} come from replacing the large logarithmic coordinate X by $-T$.

A common trap

For $x < 0$, writing simply $\log x$ hides a branch choice. The real formula [Equation \(7.14\)](#) avoids that ambiguity by using the positive quantity $T = \log(1/(-x))$. Complex branch formulas should instead retain $X = \text{Log } z + 2\pi ik$ explicitly.

7.10 The family $y + \alpha \log y = X$

Consider

$$y + \alpha \log y = X, \quad \alpha \neq 0. \quad (7.16)$$

Exponentiating gives $y^\alpha e^y = e^X$, and an exact solution is

$$y = \alpha W\left(\frac{e^{X/\alpha}}{\alpha}\right), \quad (7.17)$$

with compatible branches. Direct reversion with $h(X) = \alpha \log X$ gives a particularly simple scaling law:

$$y = X - \alpha L + \sum_{n \geq 1} \frac{\alpha^{n+1} P_n(L)}{X^n}. \quad (7.18)$$

The first terms are

$$y = X - \alpha L + \frac{\alpha^2 L}{X} + \frac{\alpha^3 L(L-2)}{2X^2} + \frac{\alpha^4 L(2L^2 - 9L + 6)}{6X^3} + \dots \quad (7.19)$$

This follows immediately from [Equation \(7.7\)](#): the m -th reversion term is multiplied by α^m .

7.11 Perturbed Lambert equations

Now consider

$$F(y) = y + \alpha \log y + \frac{\beta}{y} + \frac{\gamma \log y}{y^2} = X. \quad (7.20)$$

No new principle is required. Place

$$a(L) = \alpha L, \quad b(L) = \beta, \quad c(L) = \gamma L$$

in the universal inverse formulas of [theorem 6.2](#). The first blocks are

$$y = X - \alpha L + \frac{\alpha^2 L - \beta}{X} + \frac{\alpha^3(\frac{1}{2}L^2 - L) - \alpha\beta L + \alpha\beta - \gamma L}{X^2} + O_q(q^3). \quad (7.21)$$

This illustrates the practical value of universal reversal formulas: once the outer coefficient functions are identified, no fresh composition derivation is needed.

7.12 Branch-aware use in practice

For complex computation, proceed in this order:

1. choose the Lambert branch index k ;
2. set $X = \text{Log } z + 2\pi i k$ using a specified base branch of Log ;
3. choose the compatible branch of $L = \log X$;
4. evaluate the polynomial–logarithmic truncation;
5. if desired, apply Newton or Halley iteration without crossing a branch cut.

The formal coefficients are branch-independent; branch dependence enters through X and L .

Chapter summary

Lambert W is the compositional inverse of $Y + \log Y$ after setting $X = \text{Log } z + 2\pi i k$. Its coefficients satisfy a triangular fixed-point recurrence, an all-orders differential formula, and the closed Stirling-number identity

$$P_n(L) = \sum_{m=1}^n (-1)^{n+m} \left[\begin{matrix} n \\ n-m+1 \end{matrix} \right] \frac{L^m}{m!}.$$

The same polynomials control every equation $y + \alpha \log y = X$, with a factor α^{n+1} . Real W_{-1} near zero follows by replacing the large coordinate with $-T$. Residuals and Newton corrections provide both formal precision doubling and practical numerical refinement.

Closure, nested logarithms, and multiseries

8.1 No single small ring is closed under everything

The core ring $\mathbb{K}[L]((q))$ is deliberately economical. Its limitations are mathematically informative:

- dividing by a nonconstant leading polynomial requires rational functions of L ;
- composing with an inner factor L^β creates $\log L$;
- exponentiating a nonlinear polynomial such as L^2 creates a new transmonomial;
- reversing a leading term $X^p L^\beta$ creates fractional powers and usually $\log L$ corrections.

Rather than declaring these operations illegal, one builds a controlled hierarchy of larger classes.

8.2 A practical closure ladder

Let

$$L_1 = \log X, \quad L_2 = \log L_1, \quad \dots, \quad L_{r+1} = \log L_r.$$

A useful ladder is

$$\begin{aligned} \mathbb{K}[L_1]((X^{-1})) &\subset \mathbb{K}(L_1)((X^{-1})) \\ &\subset \mathbb{K}(L_1, L_2)((X^{-1})) \\ &\subset \mathbb{K}(L_1, L_2, L_3)((X^{-1})) \subset \dots \end{aligned} \tag{8.1}$$

For finer asymptotic expansions, replace the rational-function coefficient field by generalized Laurent or Hahn series in inverse powers of the successive logs. A typical finite-depth logarithmic monomial is

$$X^\rho L_1^{\alpha_1} L_2^{\alpha_2} \dots L_r^{\alpha_r}. \tag{8.2}$$

The dominance order reads from left to right: a change in the power of X dominates every finite change in the powers of $L_1, L_2, \dots$; at equal X -power, a change in L_1 dominates every lower logarithmic level, and so on.

Scope and closure

The fields in Equation (8.1) are convenient coefficient models, not a complete definition of logarithmic transseries. Full closure requires support conditions across all monomials (8.2). The general transseries literature supplies those conditions. For finite computations on a fixed grid and fixed logarithmic depth, the displayed models are usually sufficient.

8.3 Differentiating an iterated-log tower

The derivatives are

$$\frac{dL_1}{dX} = \frac{1}{X}, \quad \frac{dL_2}{dX} = \frac{1}{XL_1}, \quad \frac{dL_3}{dX} = \frac{1}{XL_1L_2}, \quad (8.3)$$

and in general

$$\frac{dL_j}{dX} = \frac{1}{XL_1L_2 \cdots L_{j-1}}. \quad (8.4)$$

Thus a polynomial ring in $L_1, \dots, L_r$ is not closed under differentiation once $r \geq 2$; negative powers of earlier logarithms appear. The rational field $\mathbb{K}(L_1, \dots, L_r)$, or a Laurent-monomial field, repairs this defect.

For a coefficient $a(L_1, \dots, L_r)$, the chain rule becomes

$$\frac{da}{dX} = \frac{1}{X} \left(\partial_{L_1} + \frac{1}{L_1} \partial_{L_2} + \frac{1}{L_1L_2} \partial_{L_3} + \cdots + \frac{1}{L_1 \cdots L_{r-1}} \partial_{L_r} \right) a. \quad (8.5)$$

Define the tower derivation inside parentheses by $\mathcal{D}_r$. Then

$$\frac{d}{dX} (aX^{-n}) = (\mathcal{D}_r - n)a X^{-n-1},$$

which is the exact analogue of $\Delta_n = D - n$.

8.4 Composition depth

Suppose

$$G(X) = \lambda X^p L_1^\beta (1 + u).$$

Then

$$\log G = pL_1 + \beta L_2 + \log \lambda + \log(1 + u).$$

Therefore an outer coefficient polynomial in $\log Y$ becomes a polynomial in L_1 and L_2 . If the inner leading factor also contains L_2^γ , then L_3 appears. In general, taking the logarithm of a monomial of depth r can expose depth $r + 1$.

This gives a simple planning rule:

Before composition, factor the dominant inner monomial and take its logarithm. Every logarithmic level visible there must be admitted in the output coefficient class.

8.5 Changing the fast variable

An expression with two logarithmic levels may become an ordinary polynomial–logarithmic series after a change of dominant coordinate. For example,

$$\sum_{n \geq 0} \frac{P_n(\log \log X)}{(\log X)^n} \quad (8.6)$$

is polynomial–logarithmic in the new fast variable

$$T = \log X, \quad \Lambda = \log T = \log \log X.$$

Such nested expansions are often called *multiseries*: one first expands in a dominant small scale, then allows coefficients that themselves carry an asymptotic expansion in a slower scale. Algorithms for exp–log functions and their inverses automatically discover these scales [10, 12].

8.6 Case study: inverting $y/\log y$

Consider

$$X = \frac{y}{\log y}, \quad X \rightarrow +\infty. \quad (8.7)$$

This is not tangent to the identity. Rearranging,

$$\log y = \frac{y}{X}, \quad y = e^{y/X}.$$

Multiplying by $-1/X$ gives

$$-\frac{y}{X} e^{-y/X} = -\frac{1}{X},$$

so the large solution is exactly

$$y = -XW_{-1}(-X^{-1}). \quad (8.8)$$

Set

$$T = \log X, \quad \Lambda = \log T.$$

Applying Equation (7.14) with $x = -X^{-1}$ yields

$$y \sim X \left[T + \Lambda + \sum_{n \geq 1} \frac{(-1)^{n+1} P_n(\Lambda)}{T^n} \right]. \quad (8.9)$$

The first terms are

$$y = X \left(T + \Lambda + \frac{\Lambda}{T} - \frac{\Lambda(\Lambda - 2)}{2T^2} + \frac{\Lambda(2\Lambda^2 - 9\Lambda + 6)}{6T^3} + \cdots \right). \quad (8.10)$$

This famous scale— $X \log X$, then $X \log \log X$, then inverse powers of $\log X$ with polynomial $\log \log X$ coefficients—also appears in asymptotic inversion problems surrounding prime-number estimates.

Intuition

The operation is still “polynomial–logarithmic,” but the correct fast variable is $T = \log X$, not X . Choosing the wrong fast variable makes the answer look as if it falls outside the method; choosing the correct one reveals the same Lambert polynomials again.

8.7 Case study: a logarithmic leading prefactor

Suppose

$$X = y(\log y)^\beta, \quad \beta \in \mathbb{R}. \quad (8.11)$$

A first approximation is

$$y_0 = X(\log X)^{-\beta}.$$

But

$$\log y_0 = \log X - \beta \log \log X,$$

so even the next correction contains $L_2 = \log \log X$. Write

$$y = XL_1^{-\beta}(1 + u),$$

then

$$\log y = L_1 - \beta L_2 + \log(1 + u).$$

Substituting into Equation (8.11) gives

$$1 = (1 + u) \left(1 - \beta \frac{L_2}{L_1} + \frac{\log(1 + u)}{L_1} \right)^\beta. \quad (8.12)$$

Now the small parameter is L_1^{-1} , and the coefficients are polynomials in L_2 . The first balance is

$$u = \beta^2 \frac{L_2}{L_1} + O\left(\frac{L_2^2}{L_1^2}\right). \quad (8.13)$$

Thus

$$y = XL_1^{-\beta} \left(1 + \beta^2 \frac{L_2}{L_1} + \dots \right). \quad (8.14)$$

The appearance of L_2 was predictable from the logarithm of the dominant prefactor.

8.8 Rational logarithmic coefficients versus asymptotic expansion in L^{-1}

A rational function such as

$$\frac{L^2 + 1}{L - 1}$$

can be kept exactly in $\mathbb{K}(L)$, or expanded for large L :

$$\frac{L^2 + 1}{L - 1} = L + 1 + \frac{2}{L - 1} = L + 1 + \frac{2}{L} + \frac{2}{L^2} + \dots$$

These are different representations:

- the rational form is exact and finite;
- the L^{-1} expansion resolves a finer asymptotic scale but is infinite.

A multiseries implementation should not expand rational functions in L^{-1} unless the requested scale requires it. Premature expansion causes expression swell and may obscure exact cancellation.

8.9 Exponentials and genuinely new monomials

If a coefficient contains a large nonlinear function of L , exponentiation leaves every finite logarithmic tower of rational coefficients. For example,

$$e^{L^2} = X^{\log X}, \quad e^{X^{1/2}}.$$

These are new transmonomials, not coefficient functions. A full transseries construction promotes exponentials of purely large transseries to monomials and then permits well-based sums of them. Polynomial–logarithmic arithmetic is best viewed as the height-zero or low-height sector in which these larger monomials do not yet occur.

8.10 Closure tests for the four main operations

Table 8.1: Practical closure tests.

Operation	Test	Enlargement when the test fails
Multiplication	Are coefficient products in the chosen coefficient ring, and is exponent support locally finite?	Enlarge the coefficient ring or support grid.
Division	Is the leading coefficient a unit?	Pass from $\mathbb{K}[L]$ to $\mathbb{K}(L)$, or factor logarithmic monomials explicitly.
Composition	Does $\log G$ remain in the coefficient algebra, and do substituted supports remain well based?	Add iterated logs, generalized powers, or move to a full transseries class.
Reversal	Can the dominant monomial be inverted inside the chosen monomial group, and is the normalized remainder small?	Add roots, inverse log powers, iterated logs, or exponential monomials as indicated by dominant balance.

8.11 The value of a deliberately small class

A class need not be closed under every conceivable operation to be useful. The ring $\mathbb{K}[L]((q))$ has three strong virtues:

1. every coefficient operation is finite polynomial algebra;

2. q -adic truncation makes recursive algorithms elementary;
3. many important inverse expansions, including Lambert W , stay inside it exactly.

The right strategy is to begin in the smallest class that contains the problem, perform closure tests before each operation, and enlarge only when a specific new monomial is forced.

Chapter summary

Division, composition, and reversal reveal a hierarchy of necessary extensions: polynomial coefficients, rational log coefficients, inverse-log series, iterated logarithms, generalized powers, and finally exponential transmonomials. Nested expansions often become ordinary polynomial–logarithmic series after choosing the correct fast variable. The inverse of $y/\log y$, for example, is controlled by the same Lambert polynomials in the variables $T = \log X$ and $\Lambda = \log T$.

Power–log series near zero and Dulac conventions

9.1 The small-variable dictionary

Many authors study logarithmic transseries as $x \rightarrow 0^+$ rather than $X \rightarrow +\infty$. Set

$$X = x^{-1}, \quad \ell = \log \frac{1}{x} = \log X = L. \quad (9.1)$$

Then

$$X^{-n} P(\log X) = x^n P(\ell).$$

Thus the integer-exponent core ring becomes

$$\mathbb{K}[\ell]((x)). \quad (9.2)$$

The same object has simply been re-centered at zero. The notation $\ell = \log(1/x)$ is preferable to $\log x$ on the positive real axis because $\ell \rightarrow +\infty$.

9.2 General power–log and Dulac series

A common generalized form is

$$\widehat{f}(x) = \sum_{j \geq 0} x^{\lambda_j} P_j(\ell), \quad \lambda_0 < \lambda_1 < \lambda_2 < \cdots \rightarrow +\infty, \quad (9.3)$$

where the P_j are polynomials and the exponent support satisfies suitable local-finiteness conditions. Such expansions are called power–log series; in dynamical contexts, closely related series are called Dulac series. They occur as asymptotic expansions of transition and return maps near hyperbolic polycycles. More general algebras allow real powers of ℓ and iterated logs [8, 7, 9].

The dominance order is

$$x^{\lambda_1} \ell^{m_1} \succ x^{\lambda_2} \ell^{m_2}$$

when $\lambda_1 < \lambda_2$, or when the exponents are equal and $m_1 > m_2$. As at infinity, one power of x dominates every finite power of the logarithm.

9.3 Multiplication

For

$$A = \sum_{\lambda} a_{\lambda}(\ell)x^{\lambda}, \quad B = \sum_{\mu} b_{\mu}(\ell)x^{\mu},$$

the product coefficient at exponent ν is

$$c_{\nu}(\ell) = \sum_{\lambda+\mu=\nu} a_{\lambda}(\ell)b_{\mu}(\ell). \quad (9.4)$$

The support hypotheses must ensure that this sum is finite. On a finitely generated exponent grid, the calculation is the same sparse convolution used in [algorithm 3.1](#).

9.4 Differentiation near zero

Since $\ell = -\log x$,

$$\frac{d\ell}{dx} = -\frac{1}{x}.$$

Therefore

$$\frac{d}{dx}(P(\ell)x^{\lambda}) = (\lambda P(\ell) - P'(\ell))x^{\lambda-1}. \quad (9.5)$$

The shifted operator is now $\lambda - D$, with the opposite sign from the large-variable convention. Repeated differentiation gives

$$\frac{d^r}{dx^r}(P(\ell)x^{\lambda}) = \left(\prod_{j=0}^{r-1} (\lambda - j - D) \right) P(\ell)x^{\lambda-r}. \quad (9.6)$$

This formula is indispensable when using Lagrange reversion near zero.

9.5 Units and division

In $\mathbb{K}[\ell][[x]]$, a series

$$A = a_0(\ell) + a_1(\ell)x + \cdots$$

is a unit only if a_0 is a nonzero constant. The reciprocal recurrence is identical to [Equation \(4.2\)](#), with x replacing q . If a_0 is a nonconstant polynomial in ℓ , rational logarithmic coefficients are required.

For generalized leading exponent λ_0 , factor

$$A = x^{\lambda_0} a_{\lambda_0}(\ell)(1 + U).$$

The power of x is inverted directly, the logarithmic leading coefficient is handled in the chosen coefficient field, and $(1 + U)^{-1}$ is expanded geometrically.

9.6 Near-identity composition at zero

Let

$$f(y) = y + h(y), \quad h(y) = o(y),$$

and

$$g(x) = x + H(x), \quad H(x) = o(x).$$

Then $f \circ g$ is obtained by the formal Taylor series

$$f(x + H) = \sum_{r \geq 0} \frac{f^{(r)}(x)}{r!} H^r.$$

The smallness condition is now measured relative to x . For example, $H = x^2 \ell$ is small because $x \ell \rightarrow 0$.

To track the logarithmic variable directly, write

$$g = x(1 + u), \quad u = H/x = o(1).$$

Then

$$\log \frac{1}{g} = \ell - \log(1 + u). \quad (9.7)$$

The sign differs from the large-variable formula because $\ell = -\log x$. An outer coefficient $P(\log(1/y))$ must therefore be Taylor-expanded at ℓ with increment $-\log(1 + u)$.

9.7 A normal-ordered formula near zero

Suppose

$$f(y) = \sum_n f_n \left(\log \frac{1}{y} \right) y^n \quad (9.8)$$

and

$$g(x) = \lambda x(1 + u), \quad \eta = \log(1 + u).$$

Then

$$\log \frac{1}{g} = \ell - \log \lambda - \eta, \quad g^n = \lambda^n x^n e^{n\eta}.$$

Consequently,

$$f \circ g = \sum_n \lambda^n x^n : e^{-\eta(D-n)} : f_n(\ell - \log \lambda), \quad (9.9)$$

where normal ordering again means that powers of η are not differentiated. This is the zero-centered counterpart of Equation (5.6).

9.8 Series reversal near zero

The Lagrange–Bürmann formula has exactly the same form:

$$(x + h)^{[-1]}(x) = x + \sum_{m \geq 1} \frac{(-1)^m}{m!} \left(\frac{d}{dx} \right)^{m-1} h(x)^m, \quad (9.10)$$

provided $h = o(x)$ and the support conditions make each truncation finite. The differential rule Equation (9.6) turns every term into a finite polynomial calculation.

Reversing $x + ax^2 \log(1/x)$

Let

$$f(x) = x + ax^2 \ell, \quad \ell = \log(1/x).$$

Here $h(x) = ax^2 \ell$. The first Lagrange term is

$$-h = -ax^2 \ell.$$

The second is

$$\frac{1}{2} \frac{d}{dx} h^2 = \frac{a^2}{2} \frac{d}{dx} (x^4 \ell^2) = a^2 x^3 (2\ell^2 - \ell).$$

The third term begins at x-order four. Hence

$$f^{[-1]}(x) = x - ax^2 \ell + a^2 x^3 (2\ell^2 - \ell) + O(x^4 \ell^3). \quad (9.11)$$

Substitution into f cancels the x-squared and x-cubed blocks.

9.9 Composition can create an iterated logarithm

Let

$$g(x) = x^p \ell^\beta.$$

Then

$$\log \frac{1}{g(x)} = p\ell - \beta \log \ell. \quad (9.12)$$

Thus an outer polynomial in $\log(1/y)$ becomes a polynomial in ℓ and $\log \ell$. The same closure boundary seen at infinity reappears. In Dulac normal-form problems, iterated logarithms can be forced by repeated normalization and inversion; modern logarithmic transseries frameworks are designed to accommodate them.

9.10 Parabolic and hyperbolic leading behavior

A map

$$f(x) = x + o(x)$$

is called tangent to the identity or parabolic in many dynamical settings. Its inverse is again tangent to the identity, and the triangular reversion method applies. A map

$$f(x) = \lambda x + o(x), \quad \lambda \neq 0,$$

is first normalized by its linear factor. A map

$$f(x) = x^\alpha + \text{higher terms}, \quad \alpha \neq 1,$$

has strongly non-tangent leading behavior; inversion or conjugacy may require fractional exponents and iterated logs. Formal normalization results for such classes are an active part of the theory of logarithmic transseries [8, 9].

9.11 Translation errors to avoid

When moving between $X \rightarrow \infty$ and $x \rightarrow 0^+$:

1. remember that $X^{-n} = x^n$;
2. use $\ell = \log(1/x) = \log X$, not $\log x$, unless all signs are changed consistently;
3. replace the shifted derivative $D - n$ by $n - D$ for $x^n P(\ell)$;
4. reverse the sign of the logarithmic increment in composition, as in Equation (9.7);
5. translate analytic sectors: a sector at infinity maps to a punctured sector at zero under reciprocal coordinates.

Chapter summary

The large-variable and small-variable theories are equivalent under $X = 1/x$ and $\ell = \log(1/x)$. Near zero, differentiation acts by $\lambda - D$ on a block $x^\lambda P(\ell)$, and composition changes ℓ by $-\log(1+u)$. Lagrange reversion remains valid and gives explicit power-log inverses. General Dulac supports require the same local-finiteness care as generalized series at infinity, and composition with logarithmic prefactors can force iterated logs.

Algorithms and implementation

10.1 A canonical sparse representation

For a fixed coefficient domain, represent

$$A = \sum_n a_n(L) q^n$$

as a finite map

$$n \mapsto a_n(L)$$

containing only the blocks needed below the current cutoff. For example,

$$X - L + \frac{L}{X} + \frac{L(L-2)}{2X^2}$$

is stored as

$$\{-1 \mapsto 1, 0 \mapsto -L, 1 \mapsto L, 2 \mapsto L(L-2)/2\}.$$

A sparse map is preferable to a dense list when generalized exponents or large gaps occur. For the integer q-adic core, either representation works.

Every constructor and arithmetic operation should enforce four invariants:

1. coefficient expressions are normalized in the chosen domain;
2. zero coefficients are removed;
3. no exponent at or beyond the active cutoff is stored;
4. the coefficient ring is recorded, so an unintended transition from $\mathbb{K}[L]$ to $\mathbb{K}(L)$ is visible.

10.2 Precision as part of the type

A truncated formal series is not merely an expression; it is an expression plus a guarantee:

$$A_N = [A]_{<N}, \quad A = A_N + \mathcal{O}_q(q^N).$$

Every operation should compute the strongest guaranteed output precision from the input precisions. If both inputs are known modulo q^N and have nonnegative valuation, their sum and product are known modulo q^N . If negative exponents occur, product precision requires care because an unknown high block of one factor can combine with a negative block of the other.

A safe implementation either:

- stores an explicit lower valuation and requests enough guard terms from each operand; or
- uses a lazy series interface that computes coefficients on demand.

In the common near-identity calculations, the negative part is fixed and finite—usually only the block q^{-1} —so guard precision is easy to calculate.

10.3 Core pseudocode

The following routines assume integer q -exponents and exact coefficient arithmetic.

10.3.1 Multiplication

```
function multiply(A, B, N):
  C := empty map
  for (i, ai) in A:
    for (j, bj) in B:
      if i + j < N:
        C[i+j] := C.get(i+j, 0) + ai*bj
  normalize every coefficient of C
  delete zero blocks
  return C
```

10.3.2 Reciprocal of a unit

```
function reciprocal(A, N):
  require A has valuation 0
  a0 := A[0]
  require a0 is a unit in the coefficient ring
  B[0] := 1/a0
  for n from 1 to N-1:
    s := 0
    for k from 1 to n:
      s := s + A.get(k, 0)*B.get(n-k, 0)
    B[n] := -s/a0
  return normalized B
```

10.3.3 Formal logarithm

```
function loglp(U, N):
  require valuation(U) >= 1
  S := 0
  P := 1
  for m from 1 to N-1:
    P := truncate(P*U, N)
    S := S + (-1)^(m+1)*P/m
  return truncate(S, N)
```

10.3.4 Differentiation

```
function derivative_X(A, N):
  B := empty map
  for (n, p(L)) in A:
    if n+1 < N:
      B[n+1] := derivative_L(p) - n*p
  return normalized B
```

10.4 Normal-ordered composition as an algorithm

For $G = \lambda X(1 + u)$, first compute $\delta = \log(1 + u)$. Then process each outer block independently.

```
function compose_regular(F, lambda, u, N):
  delta := log1p(u, required_guard_precision)
  H := 0
  for (n, f_n(L)) in F that can contribute below N:
    phi := f_n(L + log(lambda))
    jet := phi
    delta_power := 1
    for r from 0 while n+r < N:
      term := lambda^(-n) * q^n
              * delta_power * jet / r!
      H := truncate(H + term, N)
      delta_power := truncate(delta_power*delta, N-n)
      jet := derivative_L(jet) - n*jet
  return H
```

The variable jet stores $(D - n)^r \phi$. The algorithm never constructs an ambiguous differential-operator exponential.

For a near-identity inner series

$$G = X + A_0 + A_1 q + A_2 q^2 + \cdots,$$

form

$$u = qG - 1 = A_0 q + A_1 q^2 + A_2 q^3 + \cdots \quad (10.1)$$

and call the regular composition routine with $\lambda = 1$.

10.5 Triangular and Newton reversion

The triangular algorithm is almost definitionally simple:

```
function reverse_triangular(F, N):
  G := X
  for n from 0 to N-1:
    R := truncate(compose(F, G) - X, n+1)
    r_n := coefficient(R, q^n)
    G := G - r_n * q^n
```

```
return G
```

At the end of the n -th pass, the residual vanishes below q -order $n + 1$.

A Newton implementation uses

```
function reverse_newton(F, G, N):
  R := truncate(compose(F,G) - X, N)
  J := truncate(compose(derivative_X(F),G), N)
  dG := truncate(R / J, N)
  return truncate(G - dG, N)
```

The current approximation G should already be correct to positive q -order. If it is correct modulo q^M , request the new result modulo $q^{\min(2M,N)}$.

10.6 Coefficient-domain design

The series layer should be parameterized by its coefficient domain. Useful choices include:

1. exact polynomials $\mathbb{K}[L]$;
2. rational functions $\mathbb{K}(L)$;
3. polynomials or rational functions in $L_1, \dots, L_r$;
4. sparse Laurent monomials in the logarithmic tower;
5. symbolic expressions with a canonical simplifier.

The first two choices are especially robust. A completely general symbolic expression domain is convenient but can hide zero tests, introduce accidental branch transformations, and cause expression swell.

A good interface asks the coefficient domain to provide:

$+$, $-$, $\times$, unit test, inverse of a unit, D , substitution $L \mapsto L + c$,

and a reliable zero test.

10.7 Bell-polynomial acceleration

The direct composition algorithm builds powers δ^r . If many outer blocks share the same inner argument, precompute

$$C_{m,r} = [q^m] \frac{\delta^r}{r!} = \frac{1}{m!} B_{m,r}(1!\delta_1, 2!\delta_2, \dots).$$

Then Equation (5.7) becomes a matrix-like contraction between the Bell table and shifted derivative jets. This avoids recomputing the same powers for every outer block.

For moderate N , repeated truncated multiplication is simpler. For large N , Bell recurrences, relaxed series, or fast composition algorithms become attractive.

10.8 A tested SymPy prototype

The following compact implementation covers sparse addition, multiplication, reciprocal formation, $\log(1+u)$, differentiation, regular composition, near-identity composition, triangular reversal, and Lambert polynomial generation. The exponent n in the dictionary represents $q^n = X^{-n}$. It is intended as transparent reference code rather than a high-performance library.

```

from __future__ import annotations
from typing import Dict
import sympy as sp

L = sp.Symbol("L")
Series = Dict[int, sp.Expr]

def clean(a: Series, cutoff: int | None = None) -> Series:
    out: Series = {}
    for n, p in a.items():
        if cutoff is not None and n >= cutoff:
            continue
        p = sp.cancel(sp.expand(p))
        if p != 0:
            out[n] = p
    return out

def add(a: Series, b: Series, cutoff: int) -> Series:
    out = dict(a)
    for n, p in b.items():
        if n < cutoff:
            out[n] = out.get(n, sp.S.Zero) + p
    return clean(out, cutoff)

def scale(a: Series, c: sp.Expr, cutoff: int) -> Series:
    return clean({n: c*p for n, p in a.items()}, cutoff)

def shift_q(a: Series, s: int, cutoff: int) -> Series:
    return clean({n+s: p for n, p in a.items()
                  if n+s < cutoff}, cutoff)

def mul(a: Series, b: Series, cutoff: int) -> Series:
    out: Series = {}
    for i, ai in a.items():
        for j, bj in b.items():
            n = i + j
            if n < cutoff:
                out[n] = out.get(n, sp.S.Zero) + ai*bj
    return clean(out, cutoff)

```

```

def reciprocal_unit(a: Series, cutoff: int) -> Series:
    a0 = sp.cancel(a.get(0, sp.S.Zero))
    if a0 == 0:
        raise ValueError("q^0 coefficient must be nonzero")
    b: Series = {0: 1/a0}
    for n in range(1, cutoff):
        s = sum((a.get(k, 0)*b.get(n-k, 0)
                 for k in range(1, n+1)), sp.S.Zero)
        b[n] = -s/a0
    return clean(b, cutoff)

def loglp(u: Series, cutoff: int) -> Series:
    if u and min(u) <= 0:
        raise ValueError("u must have positive q-valuation")
    out: Series = {}
    power: Series = {0: sp.S.One}
    for m in range(1, cutoff):
        power = mul(power, u, cutoff)
        c = sp.Rational((-1)**(m+1), m)
        out = add(out, scale(power, c, cutoff), cutoff)
    return clean(out, cutoff)

def derivative_x(a: Series, cutoff: int) -> Series:
    out: Series = {}
    for n, p in a.items():
        if n+1 < cutoff:
            out[n+1] = sp.diff(p, L) - n*p
    return clean(out, cutoff)

def compose_regular(f: Series, u: Series,
                   cutoff: int, lam=sp.S.One) -> Series:
    """F(lam*X*(1+u)), modulo q**cutoff."""
    delta = loglp(u, cutoff + 2)
    out: Series = {}
    for n, poly in f.items():
        poly = sp.sympify(poly)
        phi = sp.expand(poly.subs(L, L + sp.log(lam)))
        jet = phi
        delta_power: Series = {0: sp.S.One}
        for r in range(max(0, cutoff-n)):
            coeffs = {j: p*jet for j, p in delta_power.items()}
            term = shift_q(coeffs, n, cutoff)
            term = scale(term,
                        lam**(-n)/sp.factorial(r), cutoff)
            out = add(out, term, cutoff)
            delta_power = mul(delta_power, delta, cutoff-n)
        jet = sp.diff(jet, L) - n*jet

```

```

    return clean(out, cutoff)

def near_identity_u(g: Series, cutoff: int) -> Series:
    """For  $G=X+\dots$ , return  $u=G/X-1=q*G-1$ ."""
    u = shift_q(g, 1, cutoff + 1)
    u[0] = u.get(0, 0) - 1
    return clean(u, cutoff + 1)

def compose_near_identity(f: Series, g: Series,
                          cutoff: int) -> Series:
    return compose_regular(f, near_identity_u(g, cutoff),
                           cutoff)

def reverse_triangular(f: Series, cutoff: int) -> Series:
    identity: Series = {-1: sp.S.One}
    g: Series = dict(identity)
    for n in range(0, cutoff):
        fg = compose_near_identity(f, g, n+1)
        residual = add(fg, scale(identity, -1, n+1), n+1)
        g[n] = sp.expand(g.get(n, 0)-residual.get(n, 0))
        g = clean(g, cutoff)
    return g

def lambert_polynomial(n: int) -> sp.Expr:
    S1 = sp.functions.combinatorial.numbers.stirling
    return sp.expand(sum(
        (-1)**(n+m)*S1(n, n-m+1, kind=1, signed=False)
        * L**m/sp.factorial(m)
        for m in range(1, n+1)
    ))

# Formal self-test: inverse of  $F(X)=X+\log X$ .
F: Series = {-1: 1, 0: L}
G = reverse_triangular(F, 7)
expected: Series = {-1: 1, 0: -L}
expected.update({n: lambert_polynomial(n)
                  for n in range(1, 7)})
assert clean(G, 7) == clean(expected, 7)
assert add(compose_near_identity(F, G, 7), {-1: -1}, 7) == {}
assert add(compose_near_identity(G, F, 7), {-1: -1}, 7) == {}

```

The final assertions check the Stirling formula and both composition residuals through q-order six.

10.9 Testing strategy

Formal-series software should be tested by identities, not only by matching a few printed coefficients.

10.9.1 Ring identities

For random low-degree polynomial blocks and safe cutoffs, verify

$$A(B + C) = AB + AC, \quad (AB)C = A(BC).$$

Use exact coefficients so every comparison is symbolic.

10.9.2 Reciprocal and quotient residuals

Verify

$$AA^{(-1)} = 1 + O_q(q^N), \quad A(C/A) = C + O_q(q^N).$$

Include tests whose leading coefficient is a nonconstant polynomial and confirm that the polynomial-only domain rejects them while the rational domain accepts them.

10.9.3 Composition identities

Check

$$\begin{aligned} F \circ X &= F, & X \circ G &= G, \\ (F \circ G) \circ H &= F \circ (G \circ H), \end{aligned}$$

and the chain rule Equation (5.16). Associativity tests are particularly good at detecting missing guard terms.

10.9.4 Reversal identities

For a random near-identity series, compute $G = F^{[-1]}$ and check both residuals in Equation (6.16). Then perturb the highest known coefficient and confirm that the residual changes at exactly the predicted order.

10.9.5 Lambert regression tests

Generate P_n by three independent methods:

1. the fixed-point recurrence Equation (7.6);
2. the operator formula Equation (7.8);
3. the Stirling sum Equation (7.9).

Agreement gives strong protection against sign and indexing errors.

10.10 Expression swell and normalization policy

Expanding every polynomial product immediately makes zero detection reliable but may create large intermediate expressions. Factoring every expression keeps some patterns compact but makes addition and zero tests expensive. A balanced policy is:

1. keep coefficient polynomials expanded and collected in L ;
2. keep external parameters in rational factored form when possible;
3. cancel rational functions after division but avoid unnecessary partial fractions;
4. truncate before expensive normalization;
5. expose optional factorization only for presentation.

The printed Lambert polynomials, for example, are best presented with a common factor L , but an implementation may store them expanded.

10.11 High-order improvements

For hundreds or thousands of q -orders, the following improvements matter:

- Newton reciprocal and reversion instead of coefficientwise recurrences;
- FFT-based multiplication in q and in L ;
- relaxed or online power-series algorithms;
- baby-step/giant-step composition;
- memoization of shifted derivative jets;
- Bell-polynomial or exponential-generating-function recurrences;
- modular arithmetic and rational reconstruction for exact integer coefficient problems.

The foundational formulas do not change. Fast algorithms reorganize the same finite coefficient dependencies.

10.12 Reproducibility record

For a serious symbolic computation, record:

1. the formal variable, fast scale, and logarithmic tower;
2. the coefficient domain and branch conventions;
3. the support grid and truncation order;
4. whether the truncation is blockwise in q or monomialwise in q and L ;
5. the residual identity used for verification;
6. the exact software version and simplification assumptions.

This information is often more important than the final list of coefficients, because it determines what the displayed O -term actually means.

Chapter summary

A polynomial–logarithmic series can be implemented as a sparse map from q -exponents to normalized coefficient polynomials. Precision and coefficient domain must be explicit. Multiplication, reciprocal, logarithm, differentiation, normal-ordered composition, and triangular reversal all reduce to finite loops. Newton methods accelerate high-order work, while residual identities and independent Lambert polynomial formulas provide strong tests.

Extended worked calculations

11.1 A complete quotient calculation

Let

$$\begin{aligned} N &= 1 + (2L - 1)q + (L^2 + 3)q^2 + (L^3 - L)q^3 + O_q(q^4), \\ D &= 1 + (L + 1)q + (2L - 1)q^2 + L^2q^3 + O_q(q^4). \end{aligned}$$

Seek $Q = N/D = \sum_{n=0}^3 d_n q^n + O_q(q^4)$. Since the leading coefficient of D is one, Equation (4.4) becomes

$$d_n = N_n - \sum_{k=1}^n D_k d_{n-k}.$$

Thus

$$\begin{aligned} d_0 &= 1, \\ d_1 &= (2L - 1) - (L + 1) = L - 2, \\ d_2 &= (L^2 + 3) - (L + 1)(L - 2) - (2L - 1) \\ &= 6 - L, \end{aligned}$$

and

$$\begin{aligned} d_3 &= (L^3 - L) - (L + 1)(6 - L) \\ &\quad - (2L - 1)(L - 2) - L^2 \\ &= L^3 - 2L^2 - L - 8. \end{aligned}$$

Therefore

$$\frac{N}{D} = 1 + (L - 2)q + (6 - L)q^2 + (L^3 - 2L^2 - L - 8)q^3 + O_q(q^4). \quad (11.1)$$

Multiplying Equation (11.1) by D reproduces N through q-order three.

11.2 Logarithm followed by exponential

Take

$$A = 1 + Lq + (2 - L)q^2 + (L^2 + 1)q^3 + O_q(q^4).$$

Set $U = A - 1$. Since

$$\log(1 + U) = U - \frac{U^2}{2} + \frac{U^3}{3} + O_q(q^4),$$

we find

$$\begin{aligned} \log A &= Lq + \left(2 - L - \frac{L^2}{2}\right)q^2 \\ &\quad + \left(2L^2 - L - 2 + \frac{L^3}{3}\right)q^3 + O_q(q^4). \end{aligned} \tag{11.2}$$

Let the right side be V . Expanding

$$e^V = 1 + V + \frac{V^2}{2} + \frac{V^3}{6} + O_q(q^4)$$

returns the original A . This round-trip test simultaneously checks multiplication, truncation, logarithm, and exponential routines.

11.3 Composition is not commutative

Let

$$F(X) = X + a(L), \quad G(X) = X + A(L).$$

From Equations (5.11) and (5.12),

$$F \circ G = X + (a + A) + Aa'X^{-1} + O_q(q^2),$$

while

$$G \circ F = X + (a + A) + aA'X^{-1} + O_q(q^2).$$

Hence the first commutator block is

$$F \circ G - G \circ F = (Aa' - aA')X^{-1} + O_q(q^2). \tag{11.3}$$

For $a = L^2$ and $A = L$, this becomes

$$F \circ G - G \circ F = L^2X^{-1} + O_q(q^2).$$

The noncommutativity is created entirely by the change in the logarithmic argument.

Remark 11.1. Expression (11.3) resembles the Lie bracket of the vector fields $a(L)\partial_L$ and $A(L)\partial_L$. This is not an accident: groups of near-identity substitutions have infinitesimal Lie algebras, and logarithmic coefficient derivatives encode the first nonlinear interaction.

11.4 A four-block composition with nontrivial coefficients

Let

$$F(Y) = Y + (\log^2 Y - 1) + \frac{2 \log Y}{Y} + \frac{1}{Y^2} + O(Y^{-3})$$

and

$$G(X) = X + (1 - L) + \frac{L^2}{X} + O_q(q^2).$$

Thus

$$a = L^2 - 1, \quad b = 2L, \quad c = 1, \quad A = 1 - L, \quad B = L^2, \quad C = D = 0.$$

Using theorem 5.2,

$$\begin{aligned} h_0 &= L^2 - L, \\ h_1 &= -L^2 + 4L, \\ h_2 &= L^3 + 5L^2 - 7L + 4, \\ h_3 &= \frac{4}{3}L^4 - L^3 - 8L^2 + \frac{41}{3}L - 6. \end{aligned}$$

Therefore

$$\begin{aligned} F(G(X)) &= X + (L^2 - L) + \frac{-L^2 + 4L}{X} \\ &\quad + \frac{L^3 + 5L^2 - 7L + 4}{X^2} \\ &\quad + \frac{\frac{4}{3}L^4 - L^3 - 8L^2 + \frac{41}{3}L - 6}{X^3} + O_q(q^4). \end{aligned} \tag{11.4}$$

A direct expansion of $\log G$, G^{-1} , and G^{-2} gives the same answer but requires substantially more intermediate algebra.

11.5 Reversing a perturbed logarithmic map

Consider

$$F(X) = X + L + \frac{L^2 + 1}{X} + O_q(q^2). \tag{11.5}$$

Here

$$a = L, \quad b = L^2 + 1, \quad c = d = 0.$$

Substitution into theorem 6.2 gives

$$\begin{aligned} A &= -L, \\ B &= -L^2 + L - 1, \\ C &= -L^3 + \frac{7}{2}L^2 - 2L + 1, \\ D &= -2L^4 + \frac{25}{3}L^3 - \frac{21}{2}L^2 + 6L - 2. \end{aligned}$$

Hence

$$\begin{aligned}
 F^{[-1]}(X) = X - L + \frac{-L^2 + L - 1}{X} \\
 + \frac{-L^3 + \frac{7}{2}L^2 - 2L + 1}{X^2} \\
 + \frac{-2L^4 + \frac{25}{3}L^3 - \frac{21}{2}L^2 + 6L - 2}{X^3} + O_q(q^4).
 \end{aligned} \tag{11.6}$$

This example shows that the Lambert polynomials are not a universal answer by themselves: a lower-order perturbation of the original map couples into every subsequent inverse coefficient.

11.6 A constant and logarithmic shift

Let

$$F(Y) = Y + c + \alpha \log Y. \tag{11.7}$$

Set

$$M = L + \frac{c}{\alpha}$$

when $\alpha \neq 0$. Since $c + \alpha L = \alpha M$ and $dM/dX = 1/X$, the Lagrange formula gives

$$F^{[-1]}(X) = X - \alpha M + \sum_{n \geq 1} \frac{\alpha^{n+1} P_n(M)}{X^n}. \tag{11.8}$$

Equivalently,

$$F^{[-1]}(X) = X - c - \alpha L + \frac{\alpha^2(L + c/\alpha)}{X} + \dots$$

A constant in the original perturbation shifts the logarithmic polynomial argument in every inverse coefficient.

11.7 Reversing $X - \log X$

The inverse of

$$F(Y) = Y - \log Y$$

solves $u - \log u = X$, the real equation behind W_{-1} . Put $\alpha = -1$ in Equation (7.18):

$$\begin{aligned}
 F^{[-1]}(X) = X + L + \frac{L}{X} - \frac{L(L-2)}{2X^2} \\
 + \frac{L(2L^2 - 9L + 6)}{6X^3} - \frac{L(3L^3 - 22L^2 + 36L - 12)}{12X^4} + \dots
 \end{aligned} \tag{11.9}$$

Multiplying by -1 and setting $X = T$ recovers the real W_{-1} expansion in Equation (7.15).

11.8 Newton reversion in action

For $F(Y) = Y + \log Y$, start with

$$G_1 = X - L + \frac{L}{X}.$$

As calculated earlier,

$$F(G_1) - X = \left(L - \frac{L^2}{2}\right) X^{-2} + O_q(q^3).$$

Also

$$F'(G_1) = 1 + \frac{1}{G_1} = 1 + X^{-1} + O_q(q^2).$$

The Newton correction is therefore

$$\begin{aligned} -\frac{F(G_1) - X}{F'(G_1)} &= -\left(L - \frac{L^2}{2}\right) X^{-2} + O_q(q^3) \\ &= \frac{L(L-2)}{2X^2} + O_q(q^3), \end{aligned}$$

which inserts P_2/X^2 . If the residual and denominator are carried to higher order before division, the same Newton step determines several further blocks simultaneously.

11.9 A numerical inverse check

Take

$$F(y) = y + \log y + \frac{1}{y}$$

and solve $F(y) = 100$. The expansion Equation (7.21) with $\alpha = \beta = 1$, $\gamma = 0$ gives

$$y \approx X - L + \frac{L-1}{X} + \frac{\frac{1}{2}L^2 - 2L + 1}{X^2}, \quad X = 100, \quad L = \log 100.$$

Numerically this gives approximately

$$y \approx 95.43112.$$

Substituting back produces a residual of order 10^{-5} ; one Newton correction for the original scalar equation reduces it quadratically. The formal expansion supplies a high-quality branch-aware starting value, while ordinary numerical iteration supplies final precision.

11.10 A small-variable reversal checked by composition

Return to

$$f(x) = x + ax^2\ell, \quad g(x) = x - ax^2\ell + a^2x^3(2\ell^2 - \ell).$$

To verify $f(g(x))$, first note

$$\log \frac{1}{g(x)} = \ell - \log \left(1 - ax\ell + a^2x^2(2\ell^2 - \ell)\right).$$

Through x -order one inside the logarithm,

$$\log \frac{1}{g} = \ell + ax\ell + O(x^2\ell^2).$$

Also

$$g^2 = x^2 - 2ax^3\ell + O(x^4\ell^2).$$

Therefore

$$\begin{aligned} f(g) &= g + ag^2 \log(1/g) \\ &= x - ax^2\ell + a^2x^3(2\ell^2 - \ell) + ax^2\ell + a^2x^3(\ell - 2\ell^2) + O(x^4\ell^3) \\ &= x + O(x^4\ell^3). \end{aligned}$$

Every displayed correction cancels.

11.11 How to organize a hand calculation

For a nontrivial composition or reversal, use the following worksheet.

1. Write the target precision and the exact meaning of the remainder.
2. Factor the inner series by its dominant monomial.
3. Compute only the needed terms of $\delta = \log(1 + u)$.
4. Make a dependency table: which outer blocks and derivative orders can reach each q -order?
5. Keep each q -block as a polynomial in L until the end.
6. Normalize and collect after every q -order, not after every elementary multiplication.
7. Verify the final residual before simplifying the presentation.

This method is less error-prone than expanding one enormous symbolic expression and asking a computer algebra system to sort it afterward.

Chapter summary

The examples in this chapter illustrate the full workflow: quotient recurrence, logarithm/-exponential round trips, noncommutative composition, universal composition coefficients, perturbed inverse formulas, Lambert-type shifts, Newton correction, numerical refinement, and small-variable reversion. Residual checks turn each printed expansion into a verifiable calculation.

From formal series to analytic asymptotics

12.1 Two questions that should not be conflated

A formal calculation answers:

If an expansion of the chosen form exists, what must its coefficients be?

An analytic theorem answers:

Does an actual function have that expansion, on which domain, with what remainder, and on which branch?

Formal algebra often determines coefficients uniquely and efficiently, but it does not by itself prove existence, convergence, or branch validity. This chapter explains the standard bridges in the polynomial–logarithmic setting.

12.2 An ordered asymptotic scale

On the positive real axis, the monomials

$$X^{-n}L^m, \quad n \in \mathbb{Z}, \quad m \in \mathbb{N},$$

form an asymptotic scale under the lexicographic order [Equation \(2.5\)](#). More explicitly,

$$X^{-n}L^{m-1} = o(X^{-n}L^m),$$

and, for every fixed r, s ,

$$X^{-(n+1)}L^r = o(X^{-n}L^s).$$

The second fact follows from $L^{r-s}/X \rightarrow 0$. Thus a finite polynomial q-block is asymptotically well separated from the next q-block.

If logarithmic degrees grow with q-order, a blockwise statement such as [Equation \(1.4\)](#) is usually clearer than enumerating all monomials. For sectorial complex asymptotics, replace real inequalities by uniform bounds on a sector where $\log X$ is single-valued.

12.3 Addition and multiplication of actual expansions

Suppose

$$f = A_N + R_f, \quad g = B_N + R_g,$$

where A_N, B_N are finite polynomial–logarithmic sums and the remainders are controlled at the next q -order. Then

$$fg = A_N B_N + A_N R_g + B_N R_f + R_f R_g.$$

Polynomial–logarithmic factors grow more slowly than any positive power of X , so the cross terms remain at the predicted lower order after allowing a possibly larger logarithmic power in the bound. This justifies Cauchy multiplication of Poincaré-type expansions.

The formal coefficient convolution is therefore not merely symbolic: under standard uniform remainder hypotheses, it gives the analytic product expansion.

12.4 Reciprocals and quotients

Suppose

$$f(X) = a_0 + o(1), \quad a_0 \neq 0.$$

Then f is bounded away from zero for sufficiently large X , and

$$\frac{1}{f} = \frac{1}{a_0} \frac{1}{1 + (f - a_0)/a_0}.$$

The analytic geometric series is valid once $|(f - a_0)/a_0| < 1$, and its coefficient expansion agrees with the formal reciprocal recurrence. If the leading coefficient is $a_0(L)$, one needs a region in which that function is nonzero and a coefficient class that can represent $1/a_0(L)$.

A formal unit test thus mirrors the analytic condition: the leading block must not vanish in the asymptotic region.

12.5 Logarithms and powers

If $u(X) = o(1)$, then for sufficiently large positive X , or uniformly in a suitable complex sector,

$$\log(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots.$$

The analytic Taylor remainder after M terms is $O(u^{M+1})$ when u stays in a compact subset avoiding the branch cut and -1 . Since positive q -valuation implies u^M gains at least M q -orders, formal truncation and analytic Taylor truncation agree.

For $A = cX^\rho(1 + u)$, branch choices enter through $\log c$, X^ρ , and $\log X$. The small-unit coefficients are universal once those choices are fixed.

12.6 Composition with a regular inner function

Let $G(X) = \lambda X(1 + u(X))$, where $u = o(1)$, and suppose the outer function has a uniform expansion in a region containing the image of G . Then

$$\log G = L + \log \lambda + \log(1 + u)$$

and

$$G^{-n} = \lambda^{-n} X^{-n} (1 + u)^{-n}$$

have ordinary analytic Taylor expansions. Substituting them into finitely many outer blocks and estimating the outer remainder proves the corresponding finite part of Equation (5.6).

The important analytic hypotheses are geometric:

1. $G(X)$ must tend to infinity in the outer function's asymptotic domain;
2. logarithm branches must remain consistent;
3. the perturbation u must be uniformly small;
4. the outer remainder estimate must remain uniform after substitution.

Formal support control alone does not provide these domain facts.

12.7 Residual-to-error transfer for inverse problems

A formal approximation is especially valuable when its functional residual can be converted into a rigorous error bound.

Proposition 12.1 (Real residual bound). *Let F be continuously differentiable and strictly monotone on an interval containing an exact solution y_* of $F(y_*) = X$ and an approximation y_N . If*

$$|F'(y)| \geq m > 0$$

throughout the interval between y_N and y_ , then*

$$|y_N - y_*| \leq \frac{|F(y_N) - X|}{m}. \quad (12.1)$$

Proof. The mean value theorem gives

$$F(y_N) - F(y_*) = F'(\xi)(y_N - y_*)$$

for some intermediate ξ . Take absolute values and use the lower bound on $|F'|$. □

For $F(y) = y + \log y$ on $y > 0$,

$$F'(y) = 1 + \frac{1}{y} > 1.$$

Hence the absolute residual $|y_N + \log y_N - X|$ is already an upper bound for the absolute inverse error. This makes the logarithmic Lambert residual particularly convenient.

12.8 Asymptotic correctness of near-identity reversal

The next statement explains why formal reversion coefficients are inherited by an actual inverse.

Theorem 12.1 (Transfer of a formal inverse expansion). *Suppose a real C^1 function F is eventually strictly increasing and*

$$F(y) = y + o(y), \quad F'(y) = 1 + o(1) \quad (y \rightarrow +\infty).$$

Let $G_N(X)$ be a finite polynomial–logarithmic expression such that

$$F(G_N(X)) - X = O\left(X^{-N-1}L^M\right) \quad (12.2)$$

for some M , and suppose $G_N(X) \rightarrow +\infty$. Then the eventual inverse $G = F^{-1}$ satisfies

$$G(X) - G_N(X) = O\left(X^{-N-1}L^M\right). \quad (12.3)$$

Proof. Because $F'(y) = 1 + o(1)$, there is an eventual interval on which $|F'(y)| \geq 1/2$. Both $G(X)$ and $G_N(X)$ lie in that interval for large X . Apply [proposition 12.1](#) with $m = 1/2$ and use [Equation \(12.2\)](#). $\square$

This theorem reduces analytic asymptotic inversion to two tasks: prove an actual inverse exists in the domain, and prove the formal residual has the stated analytic size.

12.9 Complex inverse problems

In the complex plane, a lower bound on $|F'|$ along a line segment does not by itself guarantee global injectivity. Typical tools include:

- the analytic inverse function theorem on a chosen local domain;
- Rouché’s theorem to count a unique zero of $F(y) - X$ in a disk;
- Newton–Kantorovich estimates;
- contraction mappings for a rearranged fixed-point equation;
- sectorial continuation with explicit branch cuts.

Once a unique nearby branch is established, a residual estimate again controls the error. The formal coefficients remain the same; only domain and branch selection change.

12.10 Convergent, asymptotic, and divergent series

Three situations should be distinguished.

12.10.1 Convergent expansion

The infinite series converges to the function on a nonempty domain. The large Lambert expansion has such a domain, although one often uses it asymptotically rather than summing blindly.

12.10.2 Divergent Poincaré expansion

For every fixed N , truncation after N terms gives the claimed remainder, but the infinite series diverges. The best numerical accuracy may occur at an X -dependent optimal truncation order.

12.10.3 Transseries with exponentially small sectors

After all algebraic and logarithmic orders, the remainder may be controlled by new scales such as e^{-X} . Different analytic sectors can then differ by exponentially small terms, leading to Stokes phenomena and resurgence. These phenomena motivate full transseries but are not needed for the algebraic operations developed here.

A common trap

The label “transseries” does not imply divergence, and the label “asymptotic expansion” does not imply convergence. Convergence is a theorem about a particular function and domain, not a formal property of the notation.

12.11 Numerical evaluation strategy

For a branch-aware inverse approximation:

1. evaluate the dominant and logarithmic coordinates with sufficient guard precision;
2. sum the formal terms in an order that minimizes cancellation;
3. compute a functional residual;
4. estimate the error from a derivative bound when available;
5. apply one or two Newton or Halley corrections;
6. verify that the iteration remains on the intended branch.

At very large X , evaluating a polynomial $P_n(L)$ by Horner’s rule is stable and efficient. At moderate X , the residual is more informative than the apparent size of the last term.

12.12 Branch cuts and symbolic simplification

Identities such as

$$\log(ab) = \log a + \log b, \quad (ab)^\alpha = a^\alpha b^\alpha$$

are not globally valid for complex principal branches. Formal monomial factorization assumes a chosen logarithmic chart. A symbolic implementation should therefore distinguish:

- formal logarithms with declared additive branch constants;
- principal numerical logarithms;
- algebraic simplifications valid only under positivity assumptions.

For real $X > 0$ and positive leading constants, the distinction is often invisible. For Lambert branches it is essential.

12.13 Documenting a remainder

A useful final formula should state:

1. the limit, such as $X \rightarrow +\infty$ or $z \rightarrow \infty$ in a sector;
2. the chosen branch of every logarithm and inverse;
3. whether $\mathcal{O}_q(q^N)$ is formal or analytic;
4. any logarithmic factor hidden in an analytic $\mathcal{O}$ -bound;
5. whether the series is known to converge or is only asymptotic.

For example,

$$W_0(z) = X - L + \sum_{n=1}^N \frac{P_n(L)}{X^n} + \mathcal{O}\left(\frac{L^{N+1}}{X^{N+1}}\right),$$

with $X = \log z$, $L = \log X$, and $z \rightarrow +\infty$, is an analytic statement. The notation

$$W = X - L + \sum_{n=1}^N P_n(L)q^n + \mathcal{O}_q(q^{N+1})$$

is a formal block statement.

Chapter summary

Formal algebra determines coefficient identities; analytic asymptotics require domain, branch, and remainder control. Standard Taylor estimates justify the arithmetic operations when the small perturbations are uniformly small. Functional residuals transfer directly to inverse errors when the derivative is bounded away from zero. Near-identity inverses with $F' = 1 + o(1)$ inherit the formal remainder order. Convergence, divergence, and exponentially small completion are separate analytic possibilities.

Pitfalls, diagnostics, and a decision procedure

13.1 Confusing the Lambert variables

The large-argument Lambert expansion uses two nested coordinates:

$$X = \text{Log } z + 2\pi i k, \quad L = \log X.$$

It is not an expansion directly in $1/z$. Replacing X by z produces an approximation of the wrong scale. On the principal positive branch,

$$W(z) \sim \log z - \log \log z,$$

not $z - \log z$.

A safe notation policy is to reserve z for the original Lambert argument, X for its branch-adjusted logarithm, and L for the logarithm of X .

13.2 Treating L as independent of X

The symptoms are missing derivative terms in composition and incorrect reversion coefficients. The cure is to use

$$\frac{d}{dX} (a(L)X^{-n}) = (a' - na)X^{-n-1}$$

or to substitute $L = \log X$ before asking a computer algebra system to differentiate.

In composition, remember that

$$L \mapsto \log G(X),$$

not merely L . A correct implementation never substitutes only the q -part of an inner series.

13.3 Assuming the polynomial coefficient ring is a field

The reciprocal of $L + q$ begins with L^{-1} , so it is not in $\mathbb{K}[L]((q))$. If a division unexpectedly produces denominators in L , this may be mathematically necessary rather than a simplification failure.

Perform the unit test before recurrence:

Is the first nonzero q -coefficient a nonzero constant in the chosen coefficient ring?

If not, enlarge the domain or keep the factor unexpanded.

13.4 Truncating without guard terms

Suppose an outer series contains q^{-1} . To compute a product modulo q^N , the other factor may be needed through q -order N , not merely $N - 1$, because $q^{-1}q^N = q^{N-1}$. Similar shifts occur in composition.

The remedy is valuation arithmetic. If a factor has lowest exponent r , an unknown coefficient at order M can affect a product as low as $M + r$. Request enough input precision so that every omitted contribution lands at or beyond the output cutoff.

13.5 Using an unordered operator exponential

The compact composition symbol

$$:e^{\delta(D-n)}:$$

is normal ordered. Removing the colons changes the meaning when $\delta = \delta(L, q)$, because $D\delta \neq 0$. A naive symbolic operator exponential then creates spurious derivatives of δ .

When in doubt, expand the definition

$$\sum_{r \geq 0} \frac{\delta^r}{r!} (D - n)^r f.$$

It is unambiguous and finite at fixed q -precision.

13.6 Composing in the wrong order

$F \circ G$ means $F(G(X))$. Since composition is noncommutative, a silent reversal changes coefficients. In code, use names such as `compose(outer, inner)` and test against

$$X \circ G = G, \quad F \circ X = F.$$

For inverse verification, check both $F \circ G$ and $G \circ F$.

13.7 Forgetting the relative smallness condition

An additive perturbation H need not be numerically small; it must be small relative to the dominant inner argument. For

$$G = X - L,$$

the correction $-L$ tends to infinity in magnitude, but

$$\frac{-L}{X} \rightarrow 0.$$

Thus $G = X(1 - L/X)$ is a valid near-identity substitution. Conversely, $G = X + X/2$ has a bounded relative perturbation but not a vanishing one; it should be normalized as $(3/2)X$, not treated with a small-unit logarithm.

13.8 Missing an iterated logarithm

If the inner leading monomial contains L^β , then

$$\log G = \cdots + \beta \log L + \cdots .$$

Trying to force the result back into $\mathbb{K}[L]((q))$ will either fail or silently replace $\log L$ by an unrelated symbol. Factor the dominant inner monomial before composition and inspect its logarithm. This predicts the necessary logarithmic depth.

13.9 Misreading a formal remainder

The statement

$$A = B + O_q(q^N)$$

means coefficient agreement below q -order N . It does not by itself assert an analytic bound with a constant. Conversely,

$$A(X) - B(X) = O(X^{-N}L^M)$$

is an analytic statement and may hide a nontrivial logarithmic exponent. Always label the formal q -adic remainder distinctly.

13.10 Applying the large expansion near a branch point

The large-argument Lambert expansion is not the correct local coordinate near $z = -e^{-1}$, where W has a square-root branch point. There the natural small variable is

$$p = \sqrt{2(ez + 1)},$$

and the expansion is a Puiseux series in p , not a polynomial–logarithmic series in $1/\log z$. An asymptotic method must begin with the local dominant balance appropriate to the limit under study.

13.11 Assuming more terms always improve a numerical approximation

Even a convergent series may have nonmonotone partial errors, and a divergent asymptotic series eventually worsens. Use the functional residual, not only the last term, to judge accuracy. If the residual starts increasing, truncate earlier or switch to Newton refinement.

13.12 Losing branch constants during logarithm splitting

For complex quantities,

$$\operatorname{Log}(ab) = \operatorname{Log} a + \operatorname{Log} b + 2\pi im$$

for an integer depending on the continuation path. In Lambert calculations, those constants are precisely what distinguish branch indices. Keep $X = \operatorname{Log} z + 2\pi ik$ intact rather than simplifying it through identities that assume principal branches compose globally.

13.13 A symptom–cause–repair table

Table 13.1: Common diagnostic patterns.

Symptom	Likely cause	Repair
Unexpected L^{-1} terms in a quotient	Nonconstant leading coefficient	Use $\mathbb{K}(L)$ or factor the leading log monomial.
Inverse residual fails one order too early	Missing guard term or off-by-one cutoff	Track valuations of all negative exponents.
Composition lacks derivatives of coefficient polynomials	L treated as independent	Replace L by $\log G$ or use shifted jets.
Spurious derivatives of the inner perturbation	Ordinary instead of normal-ordered operator exponential	Use the explicit series in Equation (5.5) .
A $\log \log X$ term appears	Inner dominant monomial contains a power of $\log X$	Enlarge logarithmic depth.
Left and right inverse checks disagree	Composition order bug or insufficient precision	Test identity substitutions and add guard terms.
Numerical iteration jumps branches	Branch-inconsistent initial coordinates	Retain k , branch cuts, and continuation path explicitly.
Coefficient expressions grow explosively	Premature expansion or no truncation	Truncate early; normalize blockwise; delay optional factoring.

13.14 A decision procedure for a new problem

Given an implicit or inverse asymptotic problem, proceed as follows.

Step 1: identify the limit and dominant monomial

Determine whether the variable tends to zero, infinity, a finite regular point, or a branch point. Solve the leading balance before choosing a series class.

Step 2: choose the fast and slow coordinates

For Lambert $W(z)$ at infinity, the fast coordinate is $X = \text{Log } z$, not z . For the inverse of $y/\log y$, the correction series is fast in $T = \log X$, with slow variable $\log T$.

Step 3: predict closure

Factor dominant monomials. Inspect reciprocals and logarithms of those factors. Add rational log coefficients, roots, or iterated logs before the coefficient calculation begins.

Step 4: choose the operation

Use convolution for products, a triangular recurrence or Newton for division, normal-ordered shifted derivatives for composition, and residual lifting, Lagrange–Bürmann, or Newton for reversal.

Step 5: compute with explicit precision

Write every intermediate result with a cutoff. Derive a dependency bound so that omitted terms cannot leak below it.

Step 6: verify formally

Use a product or composition residual. Prefer an independent formula when one exists.

Step 7: justify analytically

State the domain, branch, and remainder. Transfer the residual to an error bound if possible.

Chapter summary

Most mistakes arise from one of five sources: an undersized coefficient field, forgetting that $L = \log X$, inadequate guard precision, ambiguous branch algebra, or confusion between formal and analytic remainders. A reliable workflow begins with dominant balance and coordinate choice, performs closure tests, computes in a valued algebra, verifies a residual, and only then assigns analytic meaning.

Conclusion and further directions

Polynomial–logarithmic transseries provide a remarkably effective middle ground. They retain the finite, coefficientwise algorithms of ordinary power series while encoding the slow logarithmic scales forced by asymptotic inversion. Their basic arithmetic is elementary, but composition reveals the essential transserial mechanisms: the logarithmic variable moves, shifted derivatives appear, support and branch conditions matter, and one operation can force a larger monomial hierarchy.

The central formulas may be summarized in four lines. Multiplication is

$$[q^n](AB) = \sum_{i+j=n} a_i(L)b_j(L).$$

A reciprocal unit satisfies

$$b_0 = a_0^{-1}, \quad b_n = -a_0^{-1} \sum_{k=1}^n a_k b_{n-k}.$$

Regular composition with $G = \lambda X(1 + u)$ is

$$F \circ G = \sum_n \lambda^{-n} X^{-n} : e^{\log(1+u)(D-n)} : f_n(L + \log \lambda).$$

Near-identity reversal is

$$(X + h)^{[-1]} = X + \sum_{m \geq 1} \frac{(-1)^m}{m!} \left(\frac{d}{dX} \right)^{m-1} h^m.$$

Everything else in the core theory is organized bookkeeping, closure analysis, and verification around these identities.

Lambert W shows the method at its best. Its coefficient polynomials can be generated recursively, differentially, or combinatorially; all branches share the same polynomials; the real lower branch follows by a sign change in the large coordinate; and generalized equations $y + \alpha \log y = X$ merely scale the same family. The example also teaches a broader lesson: the correct asymptotic variable may itself be a logarithm of the original argument.

Several directions extend naturally from this foundation.

1. **Automatic scale discovery.** Algorithms can infer whether the next scale should be X^{-1} , $(\log X)^{-1}$, $\log \log X$, a root, or an exponential monomial, following the exp–log multiseries tradition.

2. **Generalized supports.** Hahn-series and grid-based methods allow real exponents and several generators while preserving finite coefficient dependencies.
3. **Iteration and conjugacy.** Near-identity composition groups lead to iterative logarithms, fractional iteration, formal normal forms, and Fatou coordinates.
4. **Differential equations.** The shifted derivative operators make polynomial–logarithmic ansätze effective for resonant asymptotic ODEs.
5. **Exponentially small completion.** When algebraic–logarithmic orders are exhausted, full transseries add exponential sectors, Stokes data, and resurgent structure.
6. **Certified symbolic computation.** Exact coefficient domains, formal valuations, residual certificates, and proof assistants can turn high- order asymptotic computations into checkable mathematical artifacts.

The practical message is simple: choose the asymptotic coordinates carefully, work in the smallest class that is genuinely closed for the next operation, and make the residual part of the answer. With those habits, multiplication, division, composition, and series reversal become systematic rather than mysterious.

Bell polynomials and coefficient extraction

A.1 Definition

The partial exponential Bell polynomials $B_{n,k}(x_1, x_2, \dots)$ are defined by

$$B_{n,k}(x_1, x_2, \dots) = \sum \frac{n!}{m_1! m_2! \dots} \prod_{j \geq 1} \left(\frac{x_j}{j!} \right)^{m_j}, \quad (\text{A.1})$$

where the sum runs over all nonnegative integer sequences satisfying

$$\sum_{j \geq 1} m_j = k, \quad \sum_{j \geq 1} j m_j = n.$$

Only $x_1, \dots, x_{n-k+1}$ can occur. The conventions are

$$B_{0,0} = 1, \quad B_{n,0} = 0 \quad (n > 0).$$

Their exponential generating function is

$$\frac{1}{k!} \left(\sum_{j \geq 1} x_j \frac{t^j}{j!} \right)^k = \sum_{n \geq k} B_{n,k}(x_1, x_2, \dots) \frac{t^n}{n!}. \quad (\text{A.2})$$

This identity is exactly what is needed for powers of a q-series.

A.2 First values

The first nonzero partial Bell polynomials are

$$\begin{aligned} B_{1,1} &= x_1, & B_{2,2} &= x_1^2, & B_{3,3} &= x_1^3, \\ B_{2,1} &= x_2, & B_{3,2} &= 3x_1x_2, & B_{4,2} &= 4x_1x_3 + 3x_2^2, \\ B_{3,1} &= x_3, & B_{4,3} &= 6x_1^2x_2, & B_{4,4} &= x_1^4. \\ B_{4,1} &= x_4, \end{aligned}$$

A useful recurrence is

$$B_{n,k} = \sum_{j=1}^{n-k+1} \binom{n-1}{j-1} x_j B_{n-j,k-1}. \quad (\text{A.3})$$

It follows by singling out the block containing a distinguished labeled element in the combinatorial interpretation, or directly from the generating function.

A.3 Powers of a q-series

Let

$$U(q) = \sum_{j \geq 1} u_j q^j.$$

Set $x_j = j!u_j$ in Equation (A.2). Then

$$\frac{U(q)^k}{k!} = \sum_{n \geq k} \frac{1}{n!} B_{n,k}(1!u_1, 2!u_2, \dots) q^n.$$

Therefore

$$[q^n]U^k = \frac{k!}{n!} B_{n,k}(1!u_1, 2!u_2, \dots). \quad (\text{A.4})$$

A.4 Logarithm and exponential coefficients

For U of positive valuation,

$$\log(1 + U) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} U^k.$$

Hence

$$[q^n] \log(1 + U) = \frac{1}{n!} \sum_{k=1}^n (-1)^{k+1} (k-1)! B_{n,k}(1!u_1, 2!u_2, \dots). \quad (\text{A.5})$$

Similarly, if $V = \sum_{j \geq 1} v_j q^j$,

$$[q^n]e^V = \frac{1}{n!} \sum_{k=0}^n B_{n,k}(1!v_1, 2!v_2, \dots). \quad (\text{A.6})$$

The sum over k is the complete exponential Bell polynomial.

A.5 Composition coefficients revisited

For regular composition, let

$$\delta = \sum_{j \geq 1} \delta_j q^j.$$

The normal-ordered block

$$:e^{\delta(D-n)}:f = \sum_{r \geq 0} \frac{\delta^r}{r!} (D-n)^r f$$

has coefficient

$$[q^m] \left(:e^{\delta(D-n)} : f \right) = \frac{1}{m!} \sum_{r=0}^m B_{m,r}(1!\delta_1, 2!\delta_2, \dots) (D-n)^r f. \quad (\text{A.7})$$

After multiplying by $\lambda^{-n} q^n$ and summing over outer blocks, one recovers Equation (5.7).

A.6 Faà di Bruno's formula

Bell polynomials also express ordinary higher derivatives of a composite:

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{k=1}^n f^{(k)}(g(x)) B_{n,k}(g'(x), g''(x), \dots, g^{(n-k+1)}(x)). \quad (\text{A.8})$$

The composition formula in this article is a close formal relative: Bell polynomials collect repeated powers of the logarithmic increment, while the shifted derivative $D - n$ simultaneously handles the outer coefficient and its power prefactor.

Deriving the universal composition formulas

B.1 Inner expansions

Let

$$G = X + A + Bq + Cq^2 + Dq^3 + O_q(q^4) = X(1 + u),$$

where

$$u = Aq + Bq^2 + Cq^3 + Dq^4 + O_q(q^5).$$

The logarithmic increment is

$$\begin{aligned} \delta = Aq + \left(B - \frac{A^2}{2}\right)q^2 + \left(C - AB + \frac{A^3}{3}\right)q^3 \\ + \left(D - AC - \frac{B^2}{2} + A^2B - \frac{A^4}{4}\right)q^4 + O_q(q^5). \end{aligned} \quad (\text{B.1})$$

The reciprocal powers needed through q-order three are

$$G^{-1} = q - Aq^2 + (A^2 - B)q^3 + (-A^3 + 2AB - C)q^4 + O_q(q^5), \quad (\text{B.2})$$

$$G^{-2} = q^2 - 2Aq^3 + (3A^2 - 2B)q^4 + O_q(q^5), \quad (\text{B.3})$$

$$G^{-3} = q^3 - 3Aq^4 + O_q(q^5). \quad (\text{B.4})$$

B.2 Outer coefficient Taylor expansions

For a coefficient function a ,

$$a(L + \delta) = a + a'\delta + \frac{1}{2}a''\delta^2 + \frac{1}{6}a'''\delta^3 + O_q(q^4). \quad (\text{B.5})$$

For b, c, d , fewer derivatives are needed after multiplication by G^{-1}, G^{-2}, G^{-3} :

$$b(L + \delta) = b + b'\delta + \frac{1}{2}b''\delta^2 + O_q(q^3),$$

$$c(L + \delta) = c + c'\delta + O_q(q^2),$$

$$d(L + \delta) = d + O_q(q).$$

Substitute Equation (B.1) and collect. For example,

$$\begin{aligned} a(L + \delta) &= a + Aa'q \\ &+ \left[\left(B - \frac{A^2}{2} \right) a' + \frac{A^2}{2} a'' \right] q^2 \\ &+ \left[\left(C - AB + \frac{A^3}{3} \right) a' + A \left(B - \frac{A^2}{2} \right) a'' + \frac{A^3}{6} a''' \right] q^3 + O_q(q^4). \end{aligned}$$

B.3 Collecting the outer series

Write

$$F(G) = G + a(\log G) + b(\log G)G^{-1} + c(\log G)G^{-2} + d(\log G)G^{-3} + \dots$$

At q-order zero, only G and a contribute, giving

$$h_0 = A + a.$$

At q-order one, contributions are

$$B + Aa' + b,$$

which gives Equation (5.12). At q-order two, use Equations (B.2) and (B.3) and the Taylor expansions to obtain

$$h_2 = -\frac{1}{2}A^2a' + \frac{1}{2}A^2a'' - Ab + Ab' + Ba' + C + c.$$

At q-order three, the contributions are:

- D from G ;
- the q-cubic Taylor block of $a(L + \delta)$;
- the product of $b(L + \delta)$ with G^{-1} through q-cubic order;
- the product of $c(L + \delta)$ with G^{-2} through q-cubic order;
- d from $d(L + \delta)G^{-3}$.

Their sum simplifies to Equation (5.14). This derivation also explains the highest derivatives that can occur at each order.

B.4 Solving for the inverse

Set $h_j = 0$ successively. The equations have the schematic form

$$h_n = \text{known expression in } A, \dots, A_{n-1} + A_n + f_n,$$

so each new inverse block enters with coefficient one. This is the explicit version of lemma 6.1. Solving the first four equations produces Equations (6.6) to (6.9).

Exercises with solutions

C.1 Exercises

Exercise C.1 (Product). Compute through q-order three:

$$(1 + Lq + q^2)(1 - Lq + (L^2 + 2)q^2 + Lq^3).$$

Exercise C.2 (Reciprocal). Find $(1 + (L + 1)q + Lq^2)^{-1}$ through q-order three.

Exercise C.3 (Differentiation). Differentiate

$$A = X^2 + L^3 + \frac{L^2 - 1}{X^2}$$

blockwise using shifted derivatives.

Exercise C.4 (A logarithm). Compute

$$\log \left(1 + (L - 1)q + (L + 2)q^2 \right)$$

through q-order two.

Exercise C.5 (Composition). Let

$$F(Y) = Y + (\log Y)^2, \quad G(X) = X - L.$$

Compute $F \circ G$ through q-order two.

Exercise C.6 (Closure). Explain why $F(Y) = \log Y$ composed with $G(X) = X^2(\log X)^3$ does not remain in $\mathbb{K}[L]((X^{-1}))$. Identify the smallest new logarithmic symbol forced by the composition.

Exercise C.7 (Reversion). Use the universal formulas to reverse

$$F(X) = X + 2L + \frac{3}{X} + O_q(q^2)$$

through q-order two.

Exercise C.8 (Lagrange formula). Apply Equation (6.13) to $F(X) = X + \alpha \log X$ and prove that the coefficient of X^{-n} is $\alpha^{n+1}P_n(L)$.

Exercise C.9 (Lambert polynomial). Compute $P_4(L)$ from the Stirling formula, using

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = 1, \quad \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 6, \quad \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 11, \quad \begin{bmatrix} 4 \\ 1 \end{bmatrix} = 6.$$

Exercise C.10 (Lower Lambert branch). Starting from $u - \log u = T$, derive the first three correction blocks for u , and hence for $W_{-1}(x) = -u$ as $x \rightarrow 0^-$.

Exercise C.11 (Small-variable inverse). Find the inverse of

$$f(x) = x + ax^2 \log(1/x) + bx^3$$

through x -order three.

Exercise C.12 (Residual bound). Let $F(y) = y + \log y$ for $y > 0$. Show that any positive approximation y_N to the solution of $F(y) = X$ satisfies

$$|y_N - y_*| \leq |y_N + \log y_N - X|.$$

C.2 Solutions

Solution 1

The q coefficient vanishes. The q -squared coefficient is

$$(L^2 + 2) - L^2 + 1 = 3.$$

The q -cubic coefficient is

$$L + L(L^2 + 2) - L = L^3 + 2L.$$

Thus the product is

$$1 + 3q^2 + (L^3 + 2L)q^3 + O_q(q^4).$$

Solution 2

Let the reciprocal coefficients be b_n . Then

$$b_0 = 1, \quad b_1 = -(L + 1),$$

$$b_2 = (L + 1)^2 - L = L^2 + L + 1,$$

and

$$\begin{aligned} b_3 &= -((L + 1)b_2 + Lb_1) \\ &= -L^3 - L^2 - L - 1. \end{aligned}$$

Therefore

$$\frac{1}{1 + (L + 1)q + Lq^2} = 1 - (L + 1)q + (L^2 + L + 1)q^2 - (L^3 + L^2 + L + 1)q^3 + O_q(q^4).$$

Solution 3

The derivative of X^2 is $2X$. For L^3 , use the n-zero block:

$$\frac{d}{dX}L^3 = 3L^2X^{-1}.$$

For $(L^2 - 1)X^{-2}$, use D - 2:

$$(2L - 2(L^2 - 1))X^{-3} = (-2L^2 + 2L + 2)X^{-3}.$$

Hence

$$A' = 2X + \frac{3L^2}{X} + \frac{-2L^2 + 2L + 2}{X^3}.$$

Solution 4

Put $U = (L - 1)q + (L + 2)q^2$. Then

$$\log(1 + U) = U - \frac{1}{2}U^2 + O_q(q^3),$$

so

$$\log(1 + U) = (L - 1)q + \left(L + 2 - \frac{(L - 1)^2}{2}\right)q^2 + O_q(q^3).$$

Solution 5

Here $a(L) = L^2$, $A = -L$, and all other displayed coefficient functions vanish. The universal formulas give

$$\begin{aligned} h_0 &= L^2 - L, & h_1 &= Aa' = -2L^2, \\ h_2 &= -\frac{1}{2}A^2a' + \frac{1}{2}A^2a'' = -L^3 + L^2. \end{aligned}$$

Thus

$$F(G) = X + (L^2 - L) - \frac{2L^2}{X} + \frac{-L^3 + L^2}{X^2} + O_q(q^3).$$

Solution 6

$$\log G = 2L + 3 \log L.$$

The new symbol is $L_2 = \log L = \log \log X$. Therefore the output lies in a two-log-level coefficient class, not in $\mathbb{K}[L]((q))$.

Solution 7

Use $a = 2L$, $b = 3$, $c = 0$. Then

$$A = -2L, \quad B = aa' - b = 4L - 3.$$

Also

$$\begin{aligned} C &= \frac{1}{2}a^2a' - a(a')^2 - ab + a'b \\ &= 4L^2 - 8L - 6L + 6 \\ &= 4L^2 - 14L + 6. \end{aligned}$$

Hence

$$F^{[-1]}(X) = X - 2L + \frac{4L - 3}{X} + \frac{4L^2 - 14L + 6}{X^2} + O_q(q^3).$$

Solution 8

In Equation (6.13), the m -th term is

$$\frac{(-1)^m \alpha^m}{m!} \left(\frac{d}{dX} \right)^{m-1} L^m.$$

It has exactly q -order $m - 1$. Put $n = m - 1$ and compare with Equation (7.8); the coefficient is $\alpha^{n+1} P_n(L)$.

Solution 9

Formula (7.9) gives

$$\begin{aligned} P_4(L) &= -\begin{bmatrix} 4 \\ 4 \end{bmatrix} L + \begin{bmatrix} 4 \\ 3 \end{bmatrix} \frac{L^2}{2!} - \begin{bmatrix} 4 \\ 2 \end{bmatrix} \frac{L^3}{3!} + \begin{bmatrix} 4 \\ 1 \end{bmatrix} \frac{L^4}{4!} \\ &= -L + 3L^2 - \frac{11}{6}L^3 + \frac{1}{4}L^4 \\ &= \frac{L(3L^3 - 22L^2 + 36L - 12)}{12}. \end{aligned}$$

Solution 10

Reversion of $u - \log u = T$ is the case $\alpha = -1$:

$$u = T + L + \frac{L}{T} - \frac{L(L-2)}{2T^2} + \frac{L(2L^2-9L+6)}{6T^3} + \dots$$

Therefore $W_{-1}(x) = -u$ with $T = \log(1/(-x))$, giving Equation (7.15).

Solution 11

Let $h(x) = ax^2\ell + bx^3$. The first reversion term is

$$-h = -ax^2\ell - bx^3.$$

Only the leading part $ax^2\ell$ of h contributes to x -order three in the second term:

$$\frac{1}{2}(h^2)' = a^2x^3(2\ell^2 - \ell) + O(x^4\ell^2).$$

Thus

$$f^{[-1]}(x) = x - ax^2\ell + (a^2(2\ell^2 - \ell) - b)x^3 + O(x^4\ell^3).$$

Solution 12

Since $F'(y) = 1 + 1/y > 1$ for $y > 0$, apply proposition 12.1 with $m = 1$.

Notation and quick reference

Symbol	Meaning
X	Large asymptotic variable.
$q = X^{-1}$	Small q-adic variable.
$L = \log X$	First slow logarithmic variable.
L_j	Iterated logarithm, $L_{j+1} = \log L_j$.
$\mathcal{P}_{\mathbb{K}}$	Polynomial–logarithmic Laurent ring $\mathbb{K}[L]((q))$.
$\mathcal{R}_{\mathbb{K}}$	Rational-log coefficient field $\mathbb{K}(L)((q))$.
$\text{val}_q A$	Lowest q-exponent of a nonzero series.
$[A]_{<N}$	Truncation to q-exponents below N .
$O_q(q^N)$	Formal q-adic remainder of valuation at least N .
D	Derivative d/dL .
Δ_n	Shifted derivative $D - n$.
$A^{(-1)}$	Multiplicative inverse $1/A$.
$F^{[-1]}$	Compositional inverse of F .
$F \circ G$	Composition $F(G(X))$.
δ	Logarithmic increment $\log(1 + u)$ for $G = \lambda X(1 + u)$.
$:e^{\delta(D-n)}:$	Normal-ordered shifted Taylor operator.
$B_{n,k}$	Partial exponential Bell polynomial.
$\begin{bmatrix} n \\ k \end{bmatrix}$	Unsigned Stirling number of the first kind.
$P_n(L)$	n-th Lambert polynomial in the expansion of W .
$X = \xi_k$	Branch coordinate $\text{Log } z + 2\pi i k$ for $W_k(z)$.

Source note and further reading

This article was developed from the collection of transseries papers, lecture notes, and tutorial drafts supplied with the project, supplemented by the primary literature and authoritative reference sources listed below. The local materials were especially useful for the general Hahn–transseries viewpoint and for Edgar’s treatment of composition. The online sources were used to verify bibliographic data, branch conventions, the all-orders Lambert expansion, and the computer-algebra literature on functional inversion.

The presentation is deliberately narrower than a full construction of the field of logarithmic–exponential transseries. Its main object is the computable sector generated by inverse powers of a large variable and polynomials (or controlled extensions) in one or more logarithms. Readers who want the complete ordered differential-field theory should begin with [4, 13, 1]; readers chiefly interested in algorithmic asymptotics and inverse functions should also consult [11, 12, 10].

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